Part 8 — Designing the parts: biosensors, regulation, libraries, protein design and natural-product hosts (weeks 26 to 31)

Part 7 was about building a construct. This part is about deciding what to build: which part, with what strength; which control layer; which library, at what size; which enzyme is worth improving and by how much; and what a natural-product project costs before the first clone is made. Every one of these weeks has an arithmetic spine that a protocol leaves out, and as in part 7 the chapters are organised around it: what the method is, what number tells you in advance whether it will work, and which control makes its output mean something.

A note on the problems. The tracker refers to numbered problems for these weeks — I1, O1, L1 to L4. I have not worked from their datasets, so wherever a calculation here would have used one I have used a constructed example with its parameters printed beside it, and said so. The method transfers; the numbers will differ from the tracker's model answers, and where they do, trust the tracker's data and check my method against it.

Two results in this part were not what I expected when I computed them, and both are reported as they came out rather than tidied. In chapter 26, the baseline of a biosensor fit is determined to about 60% if you fit it the default way and to about 6% if you weight it, which is the calibration lesson of part 4 chapter 50 arriving in a new place. In chapter 28, a burden-aware model of a three-enzyme library has "everything high" as its winner at modest burden, and the winner only moves when burden is steeper — so the honest claim is not that maximal expression is wrong, but that whether it is wrong depends on a number you do not know in advance, which is the whole case for screening a library rather than reasoning to an answer.

Ch.	Week	Chapter
26	26	Genetic parts, transcription-factor biosensors and circuits
27	27	Regulatory control: how far a gene can be turned down, and which layer to use
28	28	Libraries, expression tuning and evolution
29	29	Structure prediction and computational enzyme design: literacy
30	30	Enzyme and protein engineering for pathways
31	31	*Streptomyces* and natural-product pathway engineering

Genetic parts, transcription-factor biosensors and circuits

A part's number is only worth something if it would come out the same on someone else's instrument. A sensor's dynamic range is only worth quoting with its uncertainty. A circuit's speed is set by dilution, whatever the diagram implies.

Characterising a part so the number travels

Fluorescence from a plate reader is in arbitrary units: it depends on the gain setting, the excitation and emission filters, the path length of the well, the volume, the medium's autofluorescence and the age of the lamp. Two labs measuring the same strain report numbers that differ by a factor of ten and mean the same thing. The fix, from the in vivo reference standard approach for standard biological parts, is to measure every part against a reference promoter *on the same plate* and report the ratio — relative promoter units. The instrument's gain divides out because it multiplies both numerator and denominator.

A measurement plan for three promoters that would survive a reviewer: the reference construct in at least three wells of every plate, so plate-to-plate and well-position effects show up in its spread; at least three independent colonies per promoter, not three wells of one culture, for the reason in part 2 chapter 17; all strains in the same medium, grown to the same optical density and read in exponential phase, because a promoter's output changes with growth rate and sigma-factor availability (part 1 chapter 8); a blank and an empty-vector control for background; and normalisation to cell density, preferably checked by flow cytometry so that you are reporting fluorescence *per cell* and not per millilitre.

Three ways the same promoter behaves differently somewhere else, and what to hold fixed. **Copy number**: the same promoter on a high-copy and a low-copy plasmid differs by the copy ratio, so characterise on the vector you will use or integrate. **Upstream and downstream sequence**: the sequence before the promoter can act as an enhancer or a competing promoter, and the sequence after it sets the mRNA's 5′ structure, which sets translation initiation — part 1 chapter 8's caveat that the same ribosome binding site performs differently in front of a different gene. **Host and conditions**: a different strain, medium or growth phase changes the sigma-factor pool. The practical rule is to quote strength together with the context it was measured in, and to re-measure when any of the three changes.

Detecting that a part slows growth is the burden question of part 1 chapter 9, and the cheapest test is a growth-rate comparison of the strain with and without the part, in the same plate and medium, with enough replicates to see a few per cent. A subtler and more sensitive version is a competition: mix the two strains and follow their ratio over several generations, since a 2% growth disadvantage that is invisible in a growth curve is a factor of two in the ratio after about thirty-five generations.

The architecture of a transcription-factor biosensor

Four components: a regulator that binds the metabolite, an operator it binds on DNA, a promoter whose activity the binding changes, and a reporter. In a repressor-type sensor the regulator sits on the operator and blocks the promoter until the ligand arrives and releases it; in an activator-type sensor the ligand converts the regulator into a form that recruits polymerase. Either way a metabolite concentration becomes a transcription rate becomes, after the reporter's own maturation and dilution, a fluorescence per cell.

Regulator	Ligand	Native or first built in	Worth knowing
FapR	Malonyl-CoA	*Bacillus subtilis*, then used in *E. coli* and yeast	The sensor behind dynamic control of fatty-acid and polyketide pathways; reports a pool that is tiny and tightly regulated, so it is a sensitive reporter of acetyl-CoA carboxylation
BenM	*cis,cis*-Muconate	*Acinetobacter baylyi*	Also responds to benzoate, a close analogue — an example of why a sensor's specificity table must be measured
LysG	L-Lysine	*C. glutamicum*	Responds to lysine, arginine and histidine; useful for screening lysine producers in the industrial host, with the cross-reaction as the limit
TtgR	Naringenin and other flavonoids	*Pseudomonas putida*	A multidrug-efflux repressor whose promiscuity is the reason it senses several compounds

These are recalled from the biosensor literature rather than recomputed, and the ligand ranges are exactly the kind of cell to verify against the primary paper before building on it.

The failure modes are four, and each has a test. **Leak**: reporter output with no ligand, which sets the floor and makes the dynamic range smaller than it looks — test with the sensor in a strain that cannot make the ligand. **Saturation at low**

concentration: a sensor whose half-maximal concentration is far below your product's range gives full output for every producer and sorts nothing — test across a titration. **Host-specific behaviour**: a regulator evolved in one organism may respond differently, or to different analogues, in another, so characterise in the host you will screen in. **Ligand uptake**: if the compound is not taken up, the sensor sees only what the cell itself makes, and if it is exported, the sensor sees the intracellular pool and not the titer — which is both its value and its limitation.

Fitting a dose–response, and the uncertainty that comes with it

The four-parameter Hill function, $b + (m - b)\cdot L^n/(K^n + L^n)$, has a baseline b, a maximum m, a half-maximal concentration K and a cooperativity n. I generated nineteen points over six decades of ligand with three replicates at zero, from a known function (b = 120, m = 5,200, K = 8 µM, n = 1.6) with 8% proportional noise plus a small additive term — the same noise structure as part 4 chapter 50's calibration — and fitted it two ways.

	Baseline	Maximum	K (µM)	n	Dynamic range, 5–95% interval
True value	120	5,200	8.0	1.60	43
Default `curve_fit`	160 ± 95	4,950 ± 130	7.9 ± 0.7	1.83 ± 0.27	31 (15 to 147)
Weighted by the noise model	116 ± 7.5	4,979 ± 191	7.6 ± 0.65	1.60 ± 0.09	43 (38 to 49)

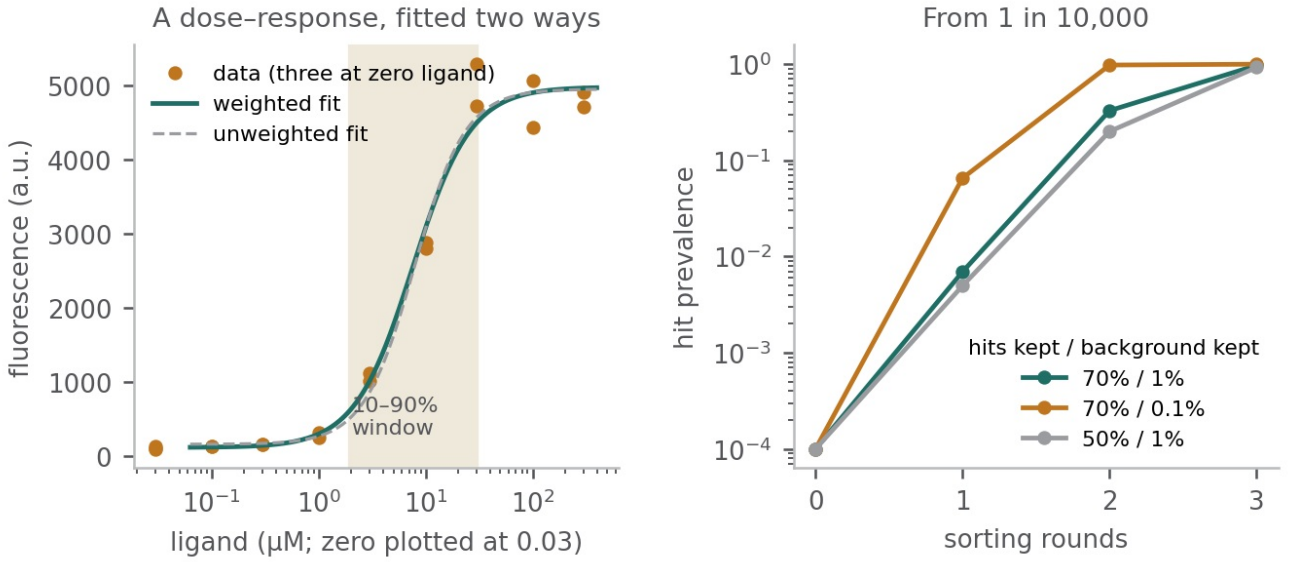


Figure 26.1 Left: the dose–response data with the weighted and unweighted fits and the 10–90% operating window shaded. The two curves are nearly indistinguishable by eye. Right: hit prevalence after one to three rounds of sorting from one in ten thousand, for three combinations of the fraction of hits kept and the fraction of background kept. Synthetic data, computed for this module; the fit parameters in the table are what the fitting routine returned, not the values I started from.

The two fits agree on K and the curve's shape and disagree by a factor of thirteen in how well the baseline is known. The reason is the lesson of chapter 50 in a new place: the default routine assumes constant noise, the high-concentration points have 8% error on a value of 5,000, and those large residuals set the scale of the uncertainty that the baseline inherits. Weighting by the noise model recovers a baseline known to about 6%. And since dynamic range is the ratio of maximum to baseline, its uncertainty is essentially the baseline's — which is why the week's phrase is correct, that dynamic range is the worst-determined of the derived quantities: it is a ratio, its denominator is the parameter pinned by the fewest points, and in the default fit it was uncertain by an order of magnitude.

Dropping the zero-ligand replicates raised the baseline's standard error by about 1.5× in the weighted fit (7.5 to 11) and left K and n essentially unchanged. The single extra measurement that would tighten the answer most is therefore **more replicates at zero and at the lowest concentrations**, since the baseline dominates the dynamic range's variance by a factor of about three over the maximum.

The operating window has a clean formula worth memorising. The ligand concentrations giving 10% and 90% of full response are $K\cdot(1/9)^{1/n}$ and $K\cdot 9^{1/n}$, so the window spans $\mathbf{81^{1/n}}$**-fold**: 81× for n = 1, 9× for n = 2, 3× for n = 4. A cooperative sensor is a switch — excellent for a sorting threshold, because a small change in metabolite produces a large change in signal, and poor as a graded readout, because it saturates within a factor of three. Choose by what the sensor is for. With the weighted fit's n = 1.6 and K = 7.6 µM the window spans 15.6-fold, from 1.9 to 30 µM.

From sensor to screen: the arithmetic of sorting

A gate sorts cells by fluorescence. Two numbers define it: the fraction of true hits that pass, r, and the fraction of background that passes, b. The enrichment per round is r/b, and the prevalence p after a round follows from Bayes: $p' = r{\cdot}p \,/\, (r{\cdot}p + b{\cdot}(1 - p))$. Starting from one hit in 10,000 with a gate keeping 70% of hits and 1% of background — 70-fold per round — prevalence is 0.7% after one round, 33% after two and 97% after three. Tighten the gate to 0.1% background and two rounds suffice.

The cost is in the other direction and is usually left out: the fraction of true hits *retained through all the rounds* is r^k, so $0.7^3 = 0.34$ — two thirds of the genuine hits are lost by the third round. The trade between purity and recovery is the same one a chromatographer makes, and the way to manage it is to sort a generous first gate, then tighten, and to keep every round's sorted population frozen so that a hit lost to a tight gate can be recovered from the one before.

Coverage and time: ten-fold coverage of a million-member library is 10^7 events, which at an assumed 20,000 events per second is 500 seconds — eight minutes. Sorting is so fast that the bottleneck is never the instrument; it is the library, the gate and the sensor, and — as part 7 chapter 24 insisted — the fact that a sorted hit is a hypothesis. The sensor reports the intracellular signal, not the titer, so every hit is confirmed by the real assay, and the confirmation rate is reported.

Circuits: a first model, and why speed costs level

The simplest circuit is a Hill-activated gene with production and loss: $dx/dt = a{\cdot}f(L) - \gamma{\cdot}x$, where f(L) is the sensor's dose–response scaled to its maximum, a is the maximal expression rate and γ is the total loss rate. The steady state is $x^* = (a/\gamma){\cdot}f(L)$: the output follows the sensor's input with a gain of a/γ. And γ is dilution plus decay. For a stable protein in a growing culture, $\gamma \approx \mu$, so the approach to steady state has a half-time of $\ln 2/\mu$ — the doubling time. At $\mu = 0.6\ h^{-1}$ that is 69 minutes, at $0.1\ h^{-1}$ it is nearly seven hours.

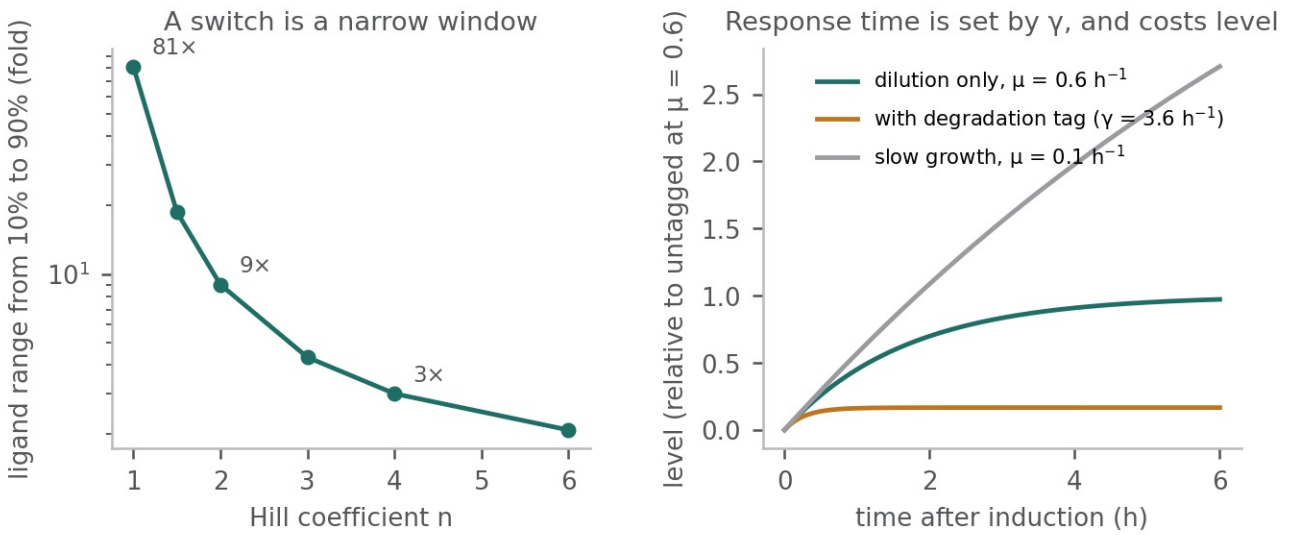


Figure 26.2 Left: the ligand range from 10% to 90% of full response against the Hill coefficient, equal to $81^{1/n}$. Right: response of a constitutively-made protein after induction, for three loss rates. A degradation tag adding $3\ h^{-1}$ cuts the half-time from 69 to 12 minutes and the steady-state level to one sixth. Computed from the first-order model in the text.

That is the engineering content of a feedback circuit and the thing a diagram hides: **a loop is as fast as its actuator can clear the protein it made**. If the control acts by dilution, the loop has a doubling-time response and a controller built on it will oscillate or lag badly. A degradation tag speeds it six-fold and costs a six-fold lower steady-state level at the same production rate — so the faster the loop, the more production capacity you must give the promoter, and the more burden you buy. There is no free speed.

Burden is the other half. A strong constitutive promoter diverts a fraction of the proteome to a protein that does not grow the cell, and part 1 chapter 9's relation says growth falls roughly in proportion, with a slope that depends on the medium. A sensor-controlled promoter wins when the protein is needed only part of the time, or only when the metabolite is present: it pays the burden in proportion to the time and level that the sensor asks for it. It loses when the sensor's own components cost more than they save, or when the loop's delay lets the cell over- and under-shoot.

Relative promoter units	`RPU = (part / reference), same plate, same conditions`
Hill response	`b + (m − b)·Lⁿ/(Kⁿ + Lⁿ); weight by the noise model`
Operating window	`L₉₀/L₁₀ = 81^(1/n); 81× at n = 1, 9× at n = 2, 3× at n = 4`

Sorting	`enrichment r/b per round; p′ = rp/(rp + b(1−p)); hits kept = r^k`
Circuit steady state	`x* = (a/γ)·f(L); half-time ln2/γ; γ = μ + k_deg`

A group reports a sensor with a 40-fold dynamic range and sorts a library with it. After two rounds the top clones show three times the fluorescence of the parent, and HPLC shows no change in titer. Give the explanations and how to separate them.

The most likely is that the sort selected the sensor rather than the product. Mutations in the reporter, in the regulator's affinity, in the plasmid's copy number or in the reporter's degradation raise fluorescence without raising the metabolite. The test is cheap: isolate the sensor plasmid from a sorted clone and put it into a fresh, unsorted parent. If fluorescence stays high, the hit was on the plasmid; if it falls back, the hit was in the chromosome and may be real. Second, a close analogue of the target may have been what the sensor saw, which is the cross-reaction column of the table above: measure the response to the analogues directly. Third, the sensor reads the intracellular pool, so a clone that has lost an exporter shows a higher signal and a lower titer — which HPLC of the broth would show as no gain, and HPLC of the pellet would show as a gain. Fourth, the gate may have enriched noise: if the single-cell distributions of parent and producer overlap heavily, sorting the top few per cent draws mostly from the parent's tail. Report the confirmation rate, because the day it falls is the day the screen stopped measuring your product.

Three things people get wrong

- ‘The sensor has a 40-fold dynamic range.’ Say with what interval: the same data gave 31 with a range of 15 to 147 and 43 with a range of 38 to 49, depending only on whether the fit used the noise model.
- ‘A steeper sensor is a better sensor.’ Steeper means a narrower window — 3× at n = 4 — which is right for a threshold and wrong for a graded readout.
- ‘The feedback loop will respond in minutes.’ Only if something removes the protein in minutes. Dilution alone gives a response time equal to the doubling time.

Self-check — answer on paper first

1. A sensor has K = 20 µM and n = 3. What ligand range does it cover between 10% and 90%, and would you use it for a graded readout?

The window spans $81^{1/3}$ = 4.3-fold, from 20 × $(1/9)^{1/3}$ = 9.6 µM to 20 × $9^{1/3}$ = 41.6 µM. That is a narrow range, so it would be a poor graded readout — a product varying over one order of magnitude would be saturated at one end and unresolved at the other — and a good threshold sensor, for example to trigger a switch or to set a sorting gate at a defined level.

2. You sort a library with a gate that keeps 60% of hits and 0.5% of background, from a hit prevalence of 10^{-5}. How many rounds, and what do you lose?

Enrichment is 60/0.5 = 120-fold per round. Prevalence goes 10^{-5} → 0.12% → 12.6% → 94%, so three rounds. Hits retained are 0.6^3 = 22%, so more than three quarters of the true hits are discarded along the way. To limit the loss, use a looser first gate and tighten in later rounds, and freeze the sorted pool after every round.

3. Why does adding a degradation tag to a feedback circuit's actuator make it faster and also weaker, and what do you do about the second?

Because the protein's loss rate γ sets both the response time, ln 2/γ, and the steady-state level, a/γ. Raising γ by tagging shortens the time and lowers the level by the same factor. To keep the same level you must raise the production rate a — a stronger promoter or ribosome binding site — which raises the burden and the leak, so the loop's speed is bought with expression capacity. The alternative is to accept a slower loop where the disturbance it must reject is itself slow.

What to open a book for. A review of transcription-factor-based biosensors for the architecture and the sensor catalogue, and Kelly et al., J. Biol. Eng. (2009) for the reference-standard method. SciPy's `curve_fit` documentation for `sigma` and `absolute_sigma` — the two arguments that turned a 95-unit baseline uncertainty into 7.5. The iGEM Registry for parts, read with the caveat that registry entries are characterised unevenly.

Regulatory control: how far a gene can be turned down, and which layer to use

Every regulator has a ceiling, and the ceiling is nearly always lower than the diagram suggests. Computing it before you build tells you whether one layer is enough or whether you must stack two.

Choosing a layer

Gene expression has levels, and part 1 chapter 8's expression equation — protein ≈ (transcription rate × translation initiation rate) / (degradation + dilution) — names every handle. Each is a control layer with its own speed, cost and ceiling.

Layer	Acts on	Speed	Typical limit	Cost and portability
Promoter, repressor	Transcription	Minutes (mRNA turns over)	Leak sets a ceiling on repression	Genome edit or plasmid; the part must work in the host
Riboswitch, sRNA	mRNA stability and translation	Minutes	Needs a degradation rate far above the native one for large knockdown	No genome edit for sRNA; works in many hosts; requires the RNA chaperone Hfq in *E. coli*
Ribosome binding site	Translation initiation	Set at construction	Context-dependent; a library is needed	A sequence change; very portable within a host
Degradation tag	Protein half-life	Sets response time	Lowers level in proportion to the added loss	Needs the host's protease; the tag must be tuned
Quorum sensing	Population density	Hours — it follows biomass	Signal is diluted by feeding; delay	Two genes plus a promoter; portable between related hosts
CRISPRi	Transcription, by blocking	Minutes to hours; protein clears by dilution	90–99% repression, not complete	One guide per target; works where editing is poor

The week's four goals, each answered with the layer and the reason. **Turn a competing pathway down, not off**: an sRNA or CRISPRi, because both give a graded reduction and neither needs a genome edit; a deletion would remove the flux entirely, which for many competing branches kills growth. **Switch production on at high density**: quorum sensing or a stationary-phase promoter, because the trigger you want is the population state. **Tune an enzyme across a ten-fold range**: a promoter or ribosome binding site library, because the target is a quantity you set once and screen, not a dynamic response. **Knock down an essential gene**: CRISPRi or an sRNA, because a knockout is not recoverable and a knockdown is.

How far can a regulator turn a gene down?

An sRNA acts by adding decay, and a knockdown of F-fold in steady-state mRNA needs the total loss rate to rise by that factor. With the native mRNA decay constant k and growth dilution μ, the extra decay needed is $(F - 1)\cdot(k + \mu)$. For a native half-life of three minutes — $k = 13.9\ h^{-1}$ — and $\mu = 0.6\ h^{-1}$:

Knockdown	Extra decay rate	Multiple of the native rate	mRNA half-life becomes
2-fold	14 h^{-1}	1×	86 s
10-fold	130 h^{-1}	9×	17 s
100-fold	1,430 h^{-1}	103×	1.7 s

A two-fold knockdown needs the sRNA to double the mRNA's decay; ten-fold needs an increase of nearly an order of magnitude and is achievable; a hundred-fold needs an mRNA half-life of under two seconds, which no degradation mechanism delivers on its own. The implication for design is the one the week asks: **do not ask a single layer for a hundred-fold knockdown**. Layers multiply, because protein level is a product of independent factors. An sRNA that cuts mRNA ten-fold and blocks translation initiation ten-fold gives a hundred-fold at the protein, from two modest mechanisms — which is in fact how most sRNAs work, since they both recruit degradation and occlude the ribosome binding site.

A repressor is limited by its leak. Even with the operator saturated, a small fraction l of promoter activity gets through, so expression relative to the unrepressed state is $l + (1 - l)/(1 + (R/K)^n)$ and the best possible repression is 1/l. A 5% leak caps repression at 20-fold, 1% at 100-fold, 0.1% at 1,000-fold. And the approach to the ceiling is slow,

because the repression curve is a Hill function. Take l = 2%, K = 10 nM and n = 2, so the ceiling is 50-fold:

Target repression	Repressor needed	As a multiple of K
2-fold	10 nM	1.0×
10-fold	34 nM	3.4×
30-fold	85 nM	8.5×
50-fold	unreachable — it is the ceiling	

Going from ten- to thirty-fold costs two and a half times the repressor, and thirty- to fifty-fold costs infinitely more. That is diminishing returns written as arithmetic, and the engineering consequence is to stack: a leaky promoter repressed to its ceiling and then weakened with a weak ribosome binding site gets a product of ceilings that no single part reaches.

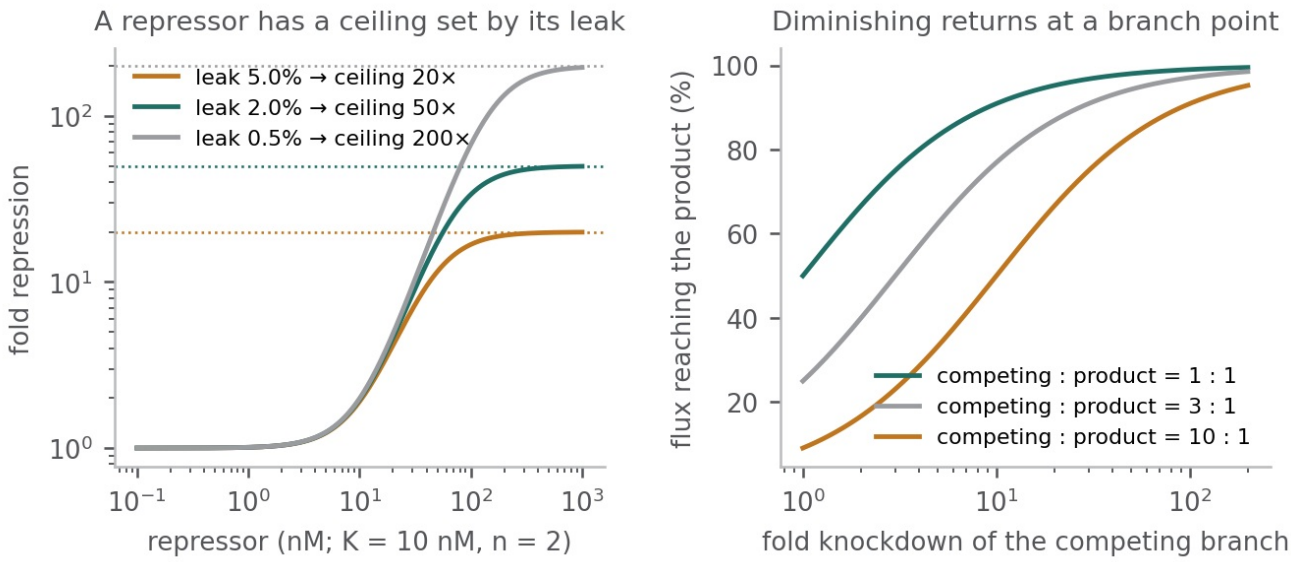


Figure 27.1 Left: fold repression against repressor amount for three leaks, with each ceiling dotted; K = 10 nM and n = 2 are assumed. Right: the fraction of flux reaching the product as a competing branch is knocked down, for three initial ratios of competing to product flux, using the first-order partition $v_p/(v_p + v_c/F)$. Computed for this module.

How much knockdown does a branch actually need?

Whether ten-fold is enough depends on how much flux the competitor takes. At a node where a metabolite is split between a product branch and a competing branch, and both enzymes are well below saturation so flux divides in proportion to activity, the fraction reaching the product after knocking the competitor down F-fold is 1/(1 + r/F), with r the initial ratio of competing to product flux.

Competing : product flux	F = 1	F = 2	F = 10	F = 100
1 : 1	50%	67%	91%	99%
3 : 1	25%	40%	77%	97%
10 : 1	9%	17%	50%	91%

Read it as a design table. When the competitor takes half, a ten-fold knockdown captures 82% of the available gain and a hundred-fold captures 98%, so the extra order of magnitude is almost worthless — ten-fold is enough, and an sRNA or a CRISPRi guide delivers it. When the competitor takes ten times the product's flux, a ten-fold knockdown only reaches half the flux and the answer is to stack layers or, if the branch is not essential, delete it. The assumptions are the ones to state alongside it: both enzymes unsaturated, no feedback, and no change in the shared pool. Part 2 chapter 15 is the reminder that the pool will move when you perturb it, and this table is an estimate to be checked against flux, not a prediction to be believed.

CRISPRi, multiplexed, and the controls it needs

Chapter 24 gave the mechanism. For several competing genes at once, a guide array expresses one guide per target and one dCas9 serves all of them, which is the multiplex repression the tracker names. The controls an experiment needs, because every one of them has caught a published result: a **non-targeting guide** in the same vector, to show the phenotype is the targeting and not the presence of dCas9 and a guide RNA; a **dCas9-only** strain, since dCas9

expression is itself a burden and some guide sequences are toxic even without a perfect genomic match; **two or more guides per target**, to exclude an off-target effect of one; an **RT-qPCR on the neighbouring genes** in the operon, to exclude polar effects; and **growth rate** of every strain, because a flux change without a growth-rate measurement may be a growth-rate change.

Dynamic control and quorum sensing: a sensor or a clock

The growth–production trade-off of part 1 chapter 9 says that expressing the pathway from the start costs growth, and a two-stage process keeps both. Quorum sensing is the autonomous version: cells secrete a small signal, it accumulates with density, and at a threshold it activates the regulator that turns the pathway on. The tracker's example controls a natural product in *S. coelicolor* this way.

The question the week asks is when to use a sensor and when a clock. **Use a sensor when the thing that should trigger the response is a metabolite** — a toxic intermediate that should throttle its own production, or a product whose accumulation should switch a pathway off. Chapter 24's loop is that. **Use a clock — a density or time trigger — when what matters is the growth phase** and the metabolic state is predictable enough that the trigger does not need to read it. The clock is simpler, cheaper and has no ligand to measure, and its weaknesses are the ones to name: the signal is diluted by a fed-batch, so the threshold density depends on the feed; at scale, mixing and gradients change how the signal accumulates; and a population under selection to grow faster will evolve strains that ignore the signal, which is the instability of part 1 chapter 10 reaching a control circuit.

A knockdown plan, with the arithmetic and the decision tree

Take a competing gene in a capstone pathway. State the layer, the fold, and why. For a non-essential competing branch that takes about as much flux as the product, ten-fold from an sRNA or a CRISPRi guide is the target, because the table above says ten-fold captures 82% of the available gain, and ten-fold needs about a nine-fold increase in mRNA decay, which is within what synthetic sRNAs are built to deliver — measure it rather than assume it.

Measure at three levels: **mRNA** by RT-qPCR, with the validated reference gene of week 22, to show the knockdown happened; **protein** by Western blot or mass spectrometry, to show it propagated, since the existing protein is cleared only by dilution; and **flux** by a carbon balance on the by-product and the product, to show it mattered. Add a pass mark in advance: for instance, at least five-fold at the mRNA, a measurable fall in the competing by-product, and a rise in product yield of at least the amount the partition table predicts, less a stated tolerance.

If flux did not move, the possibilities in order of likelihood. The enzyme you knocked down carried little control, which part 2 chapter 15 makes the commonest outcome: check its control coefficient or simply the by-product titer. The protein was still there because dilution had not cleared it: sample at a later time or after more generations. The flux re-routed through another competing branch, which the carbon balance shows as a by-product that rose while the targeted one fell. Or the knockdown reduced the enzyme but the flux was set by supply, not by the enzyme's amount, which is the saturated-enzyme case. Each has a distinct signature in the three measurements, which is the reason for taking all three.

Expression	`protein ≈ (transcription × initiation)/(degradation + dilution)`
sRNA knockdown	`extra decay = (F − 1)(k + μ); 100-fold needs ~103× the native rate`
Layers multiply	`10× at the mRNA × 10× at translation = 100× at the protein`
Repressor ceiling	`best fold = 1/leak; 5% → 20×, 1% → 100×`
Branch partition	`fraction to product = 1/(1 + r/F), r = competing : product flux`
Sensor or clock	`metabolite state → sensor; growth phase → density or time`

A colleague wants a hundred-fold knockdown of a competing gene using a single sRNA. What do you tell them, and what would you build?

That the arithmetic does not support it. A hundred-fold at steady state needs the total mRNA loss rate to rise a hundredfold, which with a native three-minute half-life means an mRNA half-life of under two seconds — beyond what the sRNA's recruitment of the degradation machinery can deliver on its own. The sRNA's real mechanism is two things at once, though: it recruits degradation and it blocks the ribosome binding site, and those multiply at the protein. So a hundred-fold at the protein is plausible from a ten-fold mRNA effect times a ten-fold translation block — which means the design question is whether this sRNA binds the ribosome binding site region, and the experiment is to measure protein and mRNA separately, since protein knockdown much larger than mRNA knockdown is the signature of the translational component.

If a hundred-fold is really needed, stack a second layer. A weaker ribosome binding site, a degradation tag on the enzyme, or a CRISPRi guide at the promoter each contribute an independent factor, and two ten-fold layers do the job with neither near its ceiling. And first check whether a hundred-fold is needed at all: if the competing branch takes half the flux, the partition table says ten-fold already captures 82% of the available gain.

Three things people get wrong

- ‘A hundred-fold knockdown needs a hundred-fold effect.’ From one mechanism, yes; layers multiply, so two ten-fold effects do it.
- ‘A stronger repressor gives more repression.’ Up to 1/leak. Past about three times K there is little left to gain, and a leaky promoter has a hard ceiling.
- ‘The knockdown worked because mRNA fell.’ The protein falls only as fast as dilution clears it, and the flux moves only if that enzyme held control.

Self-check — answer on paper first

1. A promoter has a 3% leak. What is the best repression a repressor can give, and what combination would reach 200-fold?

1/0.03 = 33-fold, whatever the repressor level. To reach 200-fold, stack a second independent layer worth at least 6-fold — a weak ribosome binding site that lowers initiation six-fold, or a degradation tag — since 33 × 6 = 200. The two ceilings multiply because transcription and translation are separate factors in the expression equation.

2. A competing branch carries four times the product's flux. You can deliver a ten-fold knockdown. What fraction of flux reaches the product, and is it enough?

1/(1 + 4/10) = 71%, up from 20% before. Whether that is enough depends on the economics, but by the diminishing-returns shape, going to a hundred-fold would add only about 25 further points — 96% — at the cost of a second layer. A sensible plan is the ten-fold knockdown, measured, with a second layer held in reserve if the measured yield falls short of the table.

3. When would you choose quorum sensing over a biosensor, and what is the first thing that can go wrong at scale?

Quorum sensing when the right moment to switch is a growth state — enough biomass has accumulated — and the metabolic state is predictable, so no metabolite needs to be read. A biosensor when the trigger is the metabolite itself. At scale the first thing to go wrong is the threshold: the signal is produced per cell and diluted by the feed, so the density at which a fed-batch crosses it differs from the one measured in a batch flask, and the switching time moves with it. Verify it by measuring signal concentration, not only reporter output, in the production vessel.

What to open a book for. Yoo et al., Nat. Biotechnol. (2013) for synthetic sRNAs in metabolic engineering, and the 2017 Microb. Cell Fact. paper on CRISPRi-guided multiplex repression, both named by the tracker — read the abstracts for which goal each was built for, then come back to the arithmetic here. The tracker's literature searches cover quorum sensing in *S. coelicolor* and CRISPRi in *V. natriegens*.

Libraries, expression tuning and evolution

When you cannot predict the optimum you search for it, and the quality of a search is decided before the first plate: by how the library is built, what the screen measures, and how many clones the arithmetic says you need.

Why a library, rather than a calculation

Part 2 chapter 15 established that control is distributed, so the best pathway is a combination of changes and not a single gene's overexpression; part 1 chapter 8 established that the same ribosome binding site performs differently in front of a different gene, so the effect of a part cannot be looked up. Together these mean the optimal expression level of each enzyme is unknown in advance and depends on the others — which is the definition of a problem that is solved by building many variants and measuring them.

The two handles are the promoter and the ribosome binding site, and they are independent, so a library can vary either or both. Promoter libraries are built by randomising bases around the −35 and −10 elements of a constitutive promoter, giving a graded range of strengths from one construct. Ribosome binding site libraries are built by randomising the Shine–Dalgarno region and its spacing to the start codon. Their advantage over a promoter library is that they change the translation initiation rate without changing the mRNA level, so within an operon each gene can be set independently — the point of the worked example in part 1 chapter 8.

The RBS Calculator, and what to expect of it

The Salis Lab calculator models translation initiation as a free-energy problem: it sums the hybridisation of the mRNA to the 16S rRNA, the cost of the spacing between them, the energy needed to unfold the local mRNA structure around the start codon, and a standby-site contribution, and converts the total into a predicted relative initiation rate. It works in two directions. Given a sequence it *predicts* a rate; given a target rate and a coding sequence it *designs* a ribosome binding site, which is the "design three spanning a hundred-fold range" exercise of the week.

What to expect, honestly: it is useful as a ranking and as a way to generate a spread, and it is less reliable as an absolute predictor, for the reason part 1 chapter 8 gave — the downstream coding sequence changes the local folding, and the model is an approximation of that folding. The reported accuracy is on a logarithmic scale with scatter of a few-fold, and I am not going to quote a figure from memory; read the validation in the primary paper and note which benchmark it was measured on and whether your host and gene resemble it. The correct use is therefore to *design a library with predicted rates spanning the range you want, then measure*, rather than to trust a prediction.

Three sites spanning a predicted hundred-fold range for one gene: aim for predicted relative rates of roughly 1, 10 and 100, evenly spaced on a log scale, rather than 1, 50 and 100, because expression effects are multiplicative and a log-spaced set samples each order of magnitude once. Record the predicted rates and the sequences, then include a reference construct with a known rate on the same plate so the measurement can be normalised as in chapter 26.

Combining them, and what the model says about it

Three enzymes at three expression levels each is $3^3 = 27$ combinations, and 95% coverage of 27 variants needs about $3 \times 27 = 81$ clones. That is a few plates, which is why a three-enzyme pathway is an ideal first combinatorial library and a ten-enzyme pathway at the same three levels (59,049) is not. To see what such a library finds, I used the three-enzyme model of part 2 chapter 16 — with the export step held fixed — set each enzyme to 0.3×, 1× or 3× its base activity, and added a burden term: a pathway proteome of 4% at baseline, shared equally, and growth falling in proportion to the proteome fraction diverted, with a slope s. Output is pathway flux multiplied by the growth factor, relative to the baseline strain.

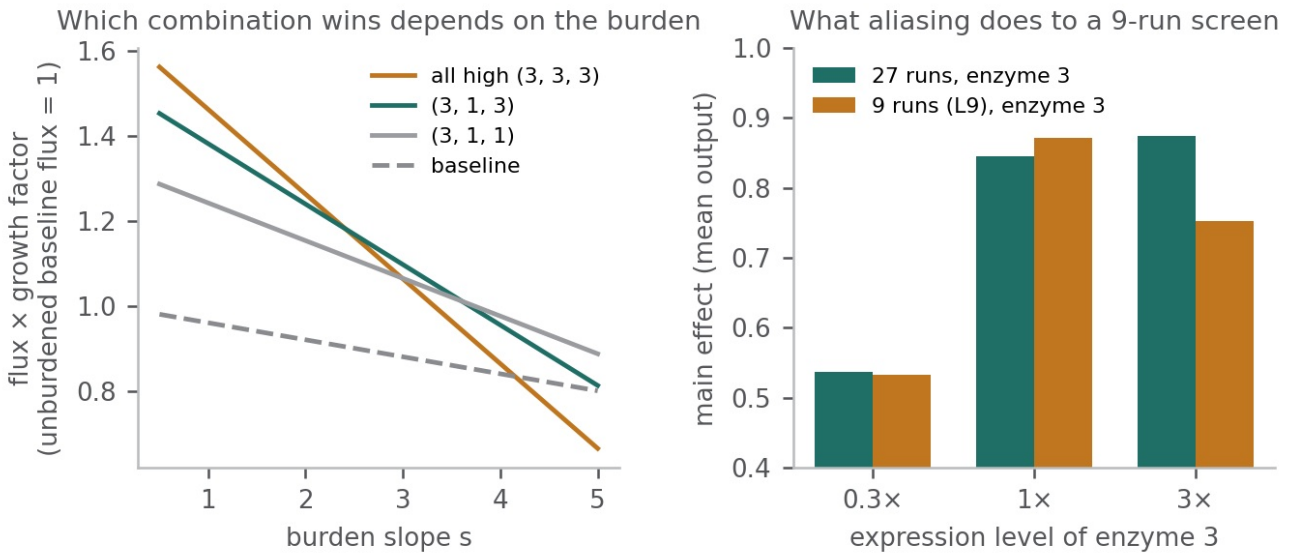


Figure 28.1 Left: output of four of the 27 combinations against the burden slope, the fraction of growth lost per unit of proteome fraction diverted; the best combination changes at about s = 2.4 and again at 3.6. Right: the effect of enzyme 3's expression level estimated from all 27 combinations and from a nine-run orthogonal array. Computed from the model of part 2 chapter 16 with the export step fixed; the enzyme levels, the 4% baseline proteome and the burden slope are assumptions.

The results, as they came out. At a burden slope of 1 — growth lost in proportion to the proteome diverted — **the best of the 27 combinations is all three enzymes at 3×**, with a net output of 1.46 against 0.96 for the baseline (which pays its burden and gains nothing). The next three are all combinations with enzyme 1 at 3×. So at modest burden, "express everything high" is correct, and my earlier warning that maximal expression is the wrong default is not supported by this model at that slope.

It becomes true as the burden steepens. Above a slope of about 2.4 the winner is (3, 1, 3): enzyme 2 held at baseline. Above about 3.6 it is (3, 1, 1), with only enzyme 1 pushed. The quantity that changes the answer is the burden slope — which part 1 chapter 9 said depends on the medium and is steeper in poor media — and nothing about the biochemistry changes. **Enzyme 1 is at its highest level in every regime**, because it carries the most control and sits before the feedback inhibition; the question is only how much of the rest of the proteome to spend on enzymes 2 and 3, which have low control and cost the same per gram.

That is the argument for a library, stated honestly: not that the optimum is surprising, but that **you cannot know the burden slope in advance**, so you cannot know which regime you are in, and building 27 variants and measuring them finds the answer without needing to. A reasoning-to-the-answer approach would have required a number that only the experiment supplies.

The second lesson is about how many runs you need. A nine-run orthogonal array — the standard L9 for three factors at three levels, a third of the full factorial — estimated enzyme 3's effect as 0.53, 0.87 and 0.75 at its three levels, a maximum at the middle level; the full 27-run design gives 0.54, 0.85 and 0.88, a maximum at the top. The array recommended (3, 3, 1) when the truth was (3, 3, 3). This is the aliasing of part 2 chapter 17 made visible: the nine-run design estimates each main effect confounded with interactions between the other factors, and in a pathway the interactions are not small because the enzymes share pools. Fractional designs are the right tool for screening many factors cheaply; for three enzymes at three levels, the full factorial is only 81 clones and there is no reason to accept the confounding.

Directed evolution

The loop is mutate, express, screen, select the best, repeat. The mutagenesis can be random — error-prone PCR, tuned to a mean number of amino-acid substitutions per gene — or focused, at positions chosen from structure or sequence conservation (chapter 30). The mutations per clone follow a Poisson distribution, which is worth tabulating because it sets the quality of a random library:

Mean mutations per gene	No change	Exactly 1	Exactly 2	3 or more
1	37%	37%	18%	8%
2	14%	27%	27%	32%
4	2%	7%	15%	76%

At two mutations per gene, 14% of the library is wild type and a third carries three or more changes, most of which will

be deleterious or neutral hitchhikers. The usual advice — aim for one to three amino-acid changes per gene — follows from this: more mutations per clone raises the rate of a beneficial mutation arriving and raises the load of harmful ones faster. And because random mutations are mostly neutral or harmful, the proportion of improved clones is small, which is why the screen must be able to find a rare positive among a large majority of negatives.

You get what you screen for. The central rule of the method is that selection acts on the measured property, exactly and only. An enzyme evolved for activity on a surrogate chromogenic substrate improves on the surrogate, and on the real substrate by whatever fraction of the improvement transfers, which may be none. An enzyme screened as a purified protein can improve in vitro and fail in the cell because its expression or solubility fell. A growth-coupled selection can be won by a strain that raised the uptake of a nutrient rather than the pathway's activity. A biosensor-based sort can be won by mutating the sensor — chapter 26's box. The defence is the same in every case: screen under conditions as close as possible to the use, on the real substrate, in the real host, and confirm every hit on the property you actually care about.

Adaptive laboratory evolution

ALE applies a selective pressure to a growing population over many generations and sequences the survivors. A daily 1:100 dilution gives $\log_2(100) = 6.6$ generations a day, so a hundred days is about 660 generations; a 1:1,000 dilution gives 10 a day and a thousand generations. It is the method of choice for phenotypes that are polygenic and hard to design — product tolerance above all, as part 2 chapter 19 argued — because evolution searches the whole genome without needing a model.

The supply arithmetic explains both its power and its limit. With a mutation rate of about 10^{-10} per base per generation — the order of magnitude for wild-type *E. coli*, to be checked against the literature for your strain — a culture of 10^{10} cells produces each of the possible single-base changes about once per generation: $N \cdot \mu = 1$ per site. So **any single mutation that helps is present almost immediately and its spread is limited by selection, not by supply**. A specific *pair* of mutations, though, arises at $N \cdot \mu^2 = 10^{-10}$ per generation, which over a thousand generations is 10^{-7} — essentially never, unless one of the pair is already established. So adaptation proceeds as a sequence of single steps, each fixed before the next arrives, which is why an evolved strain carries a handful of mutations after hundreds of generations and why a mutator, with a hundred-fold higher rate, speeds the search at the price of more passengers.

Finding the causal mutations is the real work after the evolution. Sequence the evolved population or clones from several independent lineages and look for **convergence**: the same gene or pathway mutated in different lineages is very unlikely to be coincidence. Then reconstruct each candidate in the unevolved parent, individually and in combination, because an evolved clone carries hitchhikers and the only test of causality is to put the mutation into a clean background and measure. Tolerance is "evolved rather than designed" because the causal mutations are often in global regulators and membrane functions whose connection to the phenotype is not obvious in advance — which makes ALE a hypothesis generator for the design stage rather than a replacement for it.

Selection against screening

A screen examines every candidate; a selection lets only the right ones survive. Part 1 chapter 10 made the point and the arithmetic is the whole case: a screen is limited by how many clones you can assay — 10^3 to 10^4 a day on plates, 10^7 an hour by sorting — and a selection is limited by how many cells you can plate, 10^9 or more. Coupling the product to growth is the way to make the cells do the work.

A selection that would enrich a capstone product has the same structure whatever the product: the product, or an intermediate downstream of it, becomes necessary for growth or for survival under a stress. Concretely: a transcription-factor sensor for the product drives an antibiotic resistance gene, so that producers survive a drug; or the pathway is placed so that its by-product complements an auxotrophy; or a product that detoxifies a medium component is the cell's only way to grow in it. The failure modes are the ones above, and they are specific: the cells will mutate the sensor, the resistance cassette, the transporter that admits the drug, or the copy number of the plasmid carrying it — and the defence is a second, independent confirmation and a selection stringency that is raised gradually rather than set at maximum.

Library size	`levels^enzymes; 3³ = 27 variants, ~81 clones for 95% coverage`
Spacing	`equal on a log scale: 1, 10, 100`
Poisson mutations	`P(k) = e^(−μ)μᵏ/k!; μ = 2 gives 14% wild type, 32% with three or more`
ALE generations	`log₂(dilution) per transfer; 1:100 daily = 6.6 per day`
Mutation supply	`N·μ per site; N·μ² for a specific pair; 1 and 10⁻¹⁰ at N = 10¹⁰`
Screen against selection	`10³–10⁷ clones per day against 10⁹ and above per plating`

You have a three-enzyme pathway. A colleague proposes a ten-level RBS library for each gene — 1,000 combinations — screened by HPLC at 80 samples a day. Is that sensible?

Not as stated. Ninety-five per cent coverage of 1,000 combinations is about 3,000 clones, which at 80 a day is 37 days, and that is before replicates. Two cheaper routes exist. First, three or four log-spaced levels per gene give 27 to 64 combinations and 81 to 192 clones — a week or less — and the model above says the response is smooth enough that the best of a coarse grid is near the best of a fine one. Then refine around the winner with a finer library over only the genes that mattered. Second, a proxy screen: a colorimetric or biosensor-based assay on the plate to triage the 3,000 down to the top fifty, and HPLC only on those, with the confirmation rate reported.

The deeper point is that ten levels is false precision. The measurement noise on a titer is a few per cent and the response between adjacent levels at ten levels is smaller than that, so the library is mostly samples that cannot be told apart. The sensible design uses as many levels as the assay can distinguish.

Three things people get wrong

- ‘Maximal expression is wrong.’ Not at modest burden — in the model, everything-high won up to a burden slope near 2.4. It depends on a number you do not know, which is why you screen.
- ‘A nine-run design finds the optimum.’ It found enzyme 3's best level wrongly, because main effects are confounded with interactions that pathways have.
- ‘ALE needs a high mutation rate.’ At $N = 10^{10}$ every single change is present each generation; it is the double mutants that are limiting, which is why adaptation goes step by step.

Self-check — answer on paper first

1. How many clones give 95% coverage of a library with four enzymes at four levels, and how long does it take at 200 clones a day?

$4^4 = 256$ combinations, so roughly $3 \times 256 = 770$ clones — more exactly $\ln(0.05)/\ln(1 - 1/256) \approx 765$ — which at 200 per day is about four days. That is a perfectly practical campaign. Increasing to six levels per enzyme gives 1,296 combinations and about 3,900 clones, twenty days, which is the point where a proxy screen begins to earn its keep.

2. Why can an ALE experiment find a single beneficial mutation quickly but need hundreds of generations to find a beneficial pair?

Because supply scales as $N \cdot \mu$ for a single change and $N \cdot \mu^2$ for a specific pair. At 10^{10} cells and 10^{-10} per base per generation, each single substitution appears about once per generation, but a specific double appears at 10^{-10} per generation, so it is effectively never generated from the wild-type background. A pair is found only when one member arises, spreads under selection until it is common, and then the second arises on that background — a sequence of single steps whose duration is set by how fast selection fixes each one.

3. A tolerance-evolved strain carries nine mutations. How do you decide which matter?

First sequence independent lineages and look for genes or pathways hit in more than one: convergence is strong evidence of causality. Then reconstruct candidates individually in the unevolved parent, measuring tolerance against both parent and evolved strain, and in combinations to test for interactions. A mutation that reproduces most of the phenotype alone is a driver; one that matters only in combination is a partner; one with no effect in a clean background is a passenger. Mutations in a mismatch-repair gene deserve a check of their own — a mutator phenotype explains a long list of passengers.

What to open a book for. The Salis Lab RBS Calculator documentation and its primary paper for the model and its reported accuracy; MIT OCW 7.344 on directed evolution; an ALE review, preferably one on solvent tolerance. Part 4 chapter 17 of this module is the chapter to have open for the coverage and aliasing arithmetic.

Structure prediction and computational enzyme design: literacy

A confident fold is not a confident active site, and a score with no experiment behind it is a hypothesis. This week is about knowing which tool answers which question, and what each is silent about.

What a structure prediction is, and how to read one

AlphaFold2 predicts a protein's three-dimensional structure from its sequence, using evolutionary information from a multiple sequence alignment. You rarely run it yourself: the AlphaFold database holds precomputed models for a very large number of proteins, and ColabFold gives a faster route for a sequence of your own by replacing the slow alignment step. AlphaFold3 extends the approach to complexes including small molecules, ions and nucleic acids; the terms of access for the three differ, so check the current ones before depending on any of them for a project.

Two confidence measures matter, and reading them is the first skill. **pLDDT** is a per-residue score from 0 to 100. Broadly, above 90 is very high confidence, 70 to 90 is confident for the backbone, 50 to 70 is low, and below 50 is very low and usually means the region is disordered or the model is guessing. **PAE**, the predicted aligned error, is a residue-by-residue matrix in ångströms giving the expected error in the relative position of two residues; it is what tells you whether two domains are positioned correctly relative to each other even when each is confidently folded. For a complex, **ipTM** scores the confidence in the interface.

The warning of the week, and the one most worth carrying out of it: **a high pLDDT tells you very little about the geometry that matters for catalysis.** It reports confidence in the local backbone conformation, usually to within an ångström or two. Catalysis depends on the position of side chains in the pocket to a fraction of an ångström, on the presence and placement of a cofactor or metal that the model does not contain, and on whether the enzyme is in its open or closed state. A prediction can have a pLDDT of 95 across the active site and still place a catalytic residue in a conformation the real enzyme never adopts. Treat the model as a good map of the fold and a poor measurement of the pocket.

Which tool answers which question

Question	Tool	Output	What it cannot establish
What fold does this sequence have?	AlphaFold database, ColabFold	A model with pLDDT and PAE	Ligand geometry, cofactor position, oligomeric state, conformational change
Do these two proteins associate?	AlphaFold multimer or AlphaFold3	A complex model with ipTM	Affinity; whether they associate in the cell
Design a sequence for this backbone	ProteinMPNN	Sequences predicted to fold to it	Anything involving a ligand — it sees only backbone atoms
Redesign a pocket around a cofactor or metal	LigandMPNN	Sequences conditioned on ligand atoms	Kinetics, expression, whether the ligand pose was right
Which substitutions are tolerated?	Protein language-model zero-shot scores	A score per variant	A reliable ranking — the correlation with measured activity is partial
How might a ligand sit in the pocket?	Docking	Poses and scores	A measured affinity or the true pose; scoring functions track affinity poorly
Is a structure already known?	The Protein Data Bank	An experimental structure	Look here first — an experimental structure beats any prediction for the same protein

The single distinction most student proposals get wrong. ProteinMPNN is an inverse-folding model: given a backbone, it proposes sequences likely to fold to it, and it works from backbone atoms only. It is ligand-blind. A sequence it designs for a cofactor-binding pocket has been optimised to stabilise the fold, not to hold the cofactor, and may place residues where the cofactor must sit. LigandMPNN models the atoms of cofactors, metals and small molecules explicitly, so its designs are conditioned on the ligand's presence. If your proposal redesigns a pocket and names ProteinMPNN, it has the wrong tool; if it names LigandMPNN, it has the right one, and it also has to say what the ligand coordinates came from.

Why a cofactor swap needs a ligand-aware tool

An NADPH-to-NADH swap, the engineering step part 2 chapter 19 flagged and part 5 chapter 55 applied to ketol-acid

reductoisomerase in isobutanol strains, is a pocket redesign. In Rossmann-fold enzymes the discrimination between the two cofactors is made by a small number of residues around the adenosine ribose: an NAD-specific enzyme typically has an acidic residue at the end of the second β-strand that hydrogen-bonds the 2′- and 3′-hydroxyls of the ribose, and an NADP-specific enzyme lacks it and carries basic residues that bind the 2′-phosphate. Swapping specificity means removing the basic residues, introducing the acidic one, and keeping the rest of the pocket intact — a design problem whose entire content is the cofactor's atoms, which is why a ligand-blind model cannot do it.

The redesign you would request for one step of a pathway has a fixed form. **The step**: the reduction that consumes NADPH and is the first to fail the redox balance. **The residues**: those within about 5 Å of the cofactor's adenosine ribose and 2′-phosphate, from an experimental structure or a model with the cofactor docked, since the apo prediction does not show them. **The cofactor**: modelled explicitly, NADH for the target and ideally NADPH for the counter-design. **What you measure**: the k_{cat}/K_M for each cofactor of each variant, because the aim is a change in the *ratio* without a collapse in turnover, which is the cost part 2 chapter 19 named.

Variant-effect scores: what a partial correlation is worth

Protein language models trained on sequence databases assign each substitution a score — roughly the log-likelihood of the mutant against the wild type — that correlates with measured fitness across deep-mutational-scanning data sets without any training on the enzyme in question. That is the zero-shot property. The correlation is partial: benchmark values are of the order of 0.4 to 0.5 as rank correlations, and I have not checked the current leaderboard, so look up ProteinGym rather than quoting my figure. What a partial correlation is *worth* can be computed.

Model the score and the true activity as correlated normal variables with correlation r, and ask: if you take the top 1% of variants by score, how many are in the true top 1%? Random selection gives 1%. Simulating 400,000 variants:

Correlation r	True top-1% variants in the top 1% by score	Enrichment over random
0.2	3.2%	3×
0.3	5.2%	5×
0.5	12.7%	13×
0.7	27.1%	27×

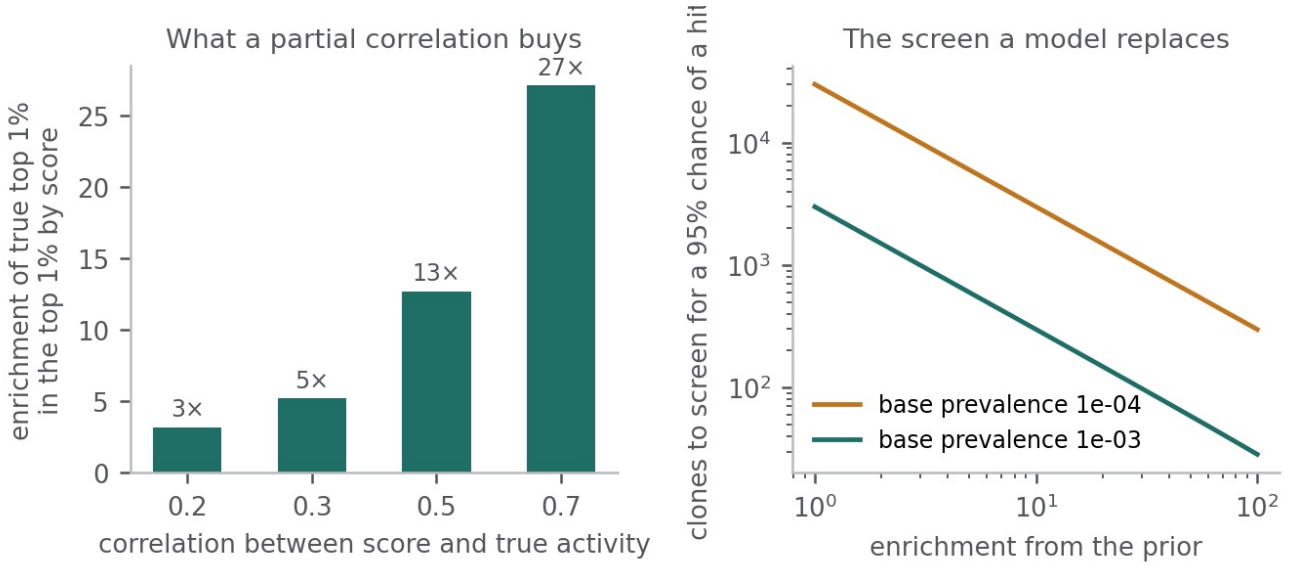


Figure 29.1 Left: enrichment of true top-1% variants in the top 1% ranked by a score with the stated correlation, from a bivariate-normal simulation. Right: the number of clones to screen for a 95% chance of at least one hit, against the enrichment a prior provides, at two base prevalences. Computed for this module; the normal model is an idealisation of real score distributions.

Now the second half. With a hit prevalence of 10^{-4}, a 95% chance of finding at least one hit takes about 30,000 clones. A prior that enriches by 13× — a correlation near 0.5 — brings that to about 2,300, and one that enriches by 100× to 300. So a score with only a partial correlation is worth an order of magnitude in screening effort (I applied an enrichment measured for the top 1% to a rarer hit class; for rarer classes the enrichment from a correlated score is generally larger, so this is on the conservative side, but simulate your own case), which is why the tracker's description of the real workflow is exactly right: **narrow a library, then still screen**. A model is a way to choose which clones to build, not a replacement for measuring them. And a high score on one variant is worth almost nothing on its own, because at r = 0.5 the variant with the top score is still more likely than not to be an unremarkable one.

Docking, and why it generates hypotheses

Docking places a ligand in a binding site and scores the pose. Poses are useful for asking which residues could contact the substrate and therefore which positions to randomise in chapter 30, and scores are useful for excluding clearly impossible placements. They are poor predictors of affinity or of catalytic competence, because scoring functions approximate solvation and entropy crudely and treat the protein as nearly rigid. The right use is to generate a short list of residues for a library, and the wrong use is to rank variants by docking score.

Writing the paragraph

The week ends with a paragraph for a proposal, and the test of it is that it names a step, a problem, a tool and a measurement. A model version: *The ketol-acid reductoisomerase step in the isobutanol pathway consumes NADPH while glycolysis supplies NADH, imposing a cofactor mismatch that costs yield (chapter 19). To shift cofactor preference we will model NADH in the experimentally determined pocket with LigandMPNN, restricting redesign to residues within 5 Å of the adenosine ribose and 2′-phosphate, and generate a library of about 200 sequences from the lowest-scoring designs. Each will be expressed in E. coli and screened in a plate assay for NADH-dependent activity against NADPH-dependent activity, selecting variants with a raised ratio and preserved k_{cat}, and the best three characterised by Michaelis–Menten kinetics with both cofactors before being returned to the production strain.*

Check it against the rule: it has the right tool, because the problem is a pocket and a cofactor; it restricts the design to a defined region, which keeps the library small; and it has an assay and a pass criterion, so the design is a plan and not a hope. A paragraph that stops at "we will use AlphaFold to redesign the enzyme" fails all three.

pLDDT bands	`>90 very high · 70–90 confident backbone · 50–70 low · <50 very low, often disordered`
PAE	`expected error in ångströms between residue pairs; domain placement`
ProteinMPNN / LigandMPNN	`backbone only (ligand-blind) / conditioned on ligand, metal and cofactor atoms`
Score enrichment	`r = 0.3 → 5×, 0.5 → 13×, 0.7 → 27× in the top 1%`
Screen size	`n = ln(0.05)/ln(1 − p·E): 30,000 at p = 10⁻⁴; 2,300 at 13×`
Workflow	`model narrows the library; the assay decides`

A student's proposal says: "We will use ProteinMPNN to redesign the cofactor-binding pocket of the reductase to accept NADH." What is wrong and what do you suggest?

The tool is blind to the thing being redesigned. ProteinMPNN sees backbone atoms only, so it has no representation of the cofactor and will propose pocket sequences optimised for stability of the fold — which can place a bulky residue where the adenosine ribose needs to sit. The right tool is LigandMPNN, or a structure-based design workflow in which the cofactor is explicit. Then three requirements the proposal does not meet: a source for the ligand coordinates, since the predicted apo model does not contain the cofactor, so an experimental structure with a cofactor or a documented docking step is needed; a limited residue set, because redesigning the whole protein is neither plausible nor testable; and an assay with a pass mark — the ratio of NADH to NADPH activity, with a floor on turnover.

I would also ask for the counter-design: a model that merely makes NADH binding better does not guarantee NADPH binding is worse, and a cofactor *swap* needs both. And I would ask what the fallback is if the designed library gives nothing, which is saturation mutagenesis of the three or four residues the literature on Rossmann-fold specificity already names — the oldest method, and the one with a track record.

Three things people get wrong

- ‘pLDDT is 95, so the active site is right.’ pLDDT is confidence in local backbone placement; the pocket geometry that sets catalysis is finer than that and the model has no cofactor.
- ‘ProteinMPNN designs enzyme pockets.’ It is ligand-blind. LigandMPNN is the tool that sees the cofactor.
- ‘The model's top-ranked variant will be the best.’ At a correlation of 0.5 it is more likely unremarkable than outstanding; the value is in enriching a library.

Self-check — answer on paper first

1. A model with a correlation of 0.3 to measured activity is used to pick the top 1% of a 10^5-variant library. Hit prevalence is 10^{-4}. How many clones must you screen with and without it?

Without the prior, $\ln(0.05)/\ln(1 - 10^{-4}) \approx 30{,}000$ clones for a 95% chance of at least one hit. With 5× enrichment the prevalence in the selected subset is 5×10^{-4}, so about 6,000. The 1,000 variants selected from 10^5 can be screened

outright, but the probability of any hit among them is then $1 - (1 - 5 \times 10^{-4})^{1000} \approx 39\%$ — so a correlation of 0.3 narrows the search but does not remove the need to screen deep.

2. Why does an AlphaFold model usually contain no information about whether your enzyme will accept an alternative substrate?

Because substrate specificity is decided by the shape and chemistry of the pocket at sub-ångström resolution and by the dynamics of the open and closed states, and the model predicts the apo backbone with a confidence that does not extend to side-chain positions in a binding site. It also has no ligand in it, so nothing says how a substrate would sit. A prediction can propose which residues line the pocket; whether a different substrate fits is a docking-and-experiment question.

3. Name the three things a computational enzyme-engineering paragraph must contain to count as a plan.

A named step with a stated problem — which reaction, and what limits it. A tool that is appropriate to the problem, correctly named, with its inputs specified: a cofactor-binding pocket needs a ligand-aware method and a source of ligand coordinates. And the experiment that tests the prediction: the assay, the variants measured, and the criterion for success. A design with no assay is a hypothesis, not a plan.

What to open a book for. Dauparas et al., Nat. Methods (2025) for LigandMPNN and the Institute for Protein Design's announcement of it; the ColabFold paper for what structure prediction does and does not give you; and the AlphaFold database entry for one enzyme in your own pathway — fetch it, read its pLDDT and PAE, and write what it cannot tell you. For variant-effect benchmarks, the ProteinGym leaderboard is the place to check the current correlations.

Enzyme and protein engineering for pathways

Before spending a campaign on an enzyme, compute what the improvement is worth. For a chain of first-order steps that calculation is exact, takes one line, and usually says the answer is smaller than the enthusiasm.

Is this enzyme worth engineering?

Part 2 chapter 15 derived the two-step case by hand: with fixed substrate and first-order steps, the flux is $J = k_1k_2S/(k_2 + k_1/K_1)$, which is a flux driven through two *resistances in series*. Write each step's resistance as $R_i \propto 1/(\text{enzyme activity})$. Then $J \propto 1/\Sigma R_i$, the control coefficient of step i is $C_i = R_i/\Sigma R$, and multiplying enzyme i's activity by m shrinks its resistance by m. The new flux relative to the old is

```
J_new / J = 1 / (1 − C_i·(1 − 1/m))        and as m → ∞:  J_max / J = 1 / (1 − C_i)
```

That is exact for a chain of first-order steps and it is the engineer's answer to "what is a faster enzyme worth". A perfect enzyme — infinite activity, zero resistance — can raise flux by at most $1/(1 - C)$, no matter how much protein engineering is spent on it.

Control coefficient	2× activity	10× activity	Ceiling (infinite)
0.05	+2.6%	+4.7%	+5.3%
0.10	+5.3%	+9.9%	+11.1%
0.30	+17.6%	+37%	+43%
0.60	+43%	+117%	+150%

Check it against a model that is not first order. Part 2 chapter 16's three-enzyme pathway has saturable steps, a reversible kinetic form and feedback, and enzyme 2 there carried a control coefficient of 0.102. The formula predicts +5.4% for a doubling, +8.9% for fivefold, +10.7% for twentyfold and a ceiling of +11.4%; the full model gave +4.8%, +7.4%, +8.7% and a plateau near 9%. The formula overstates by one to two percentage points because real control coefficients fall as you raise the enzyme and feedback absorbs part of the gain — so it is an *upper* estimate, which is the right direction to err when deciding whether a campaign is worth starting.

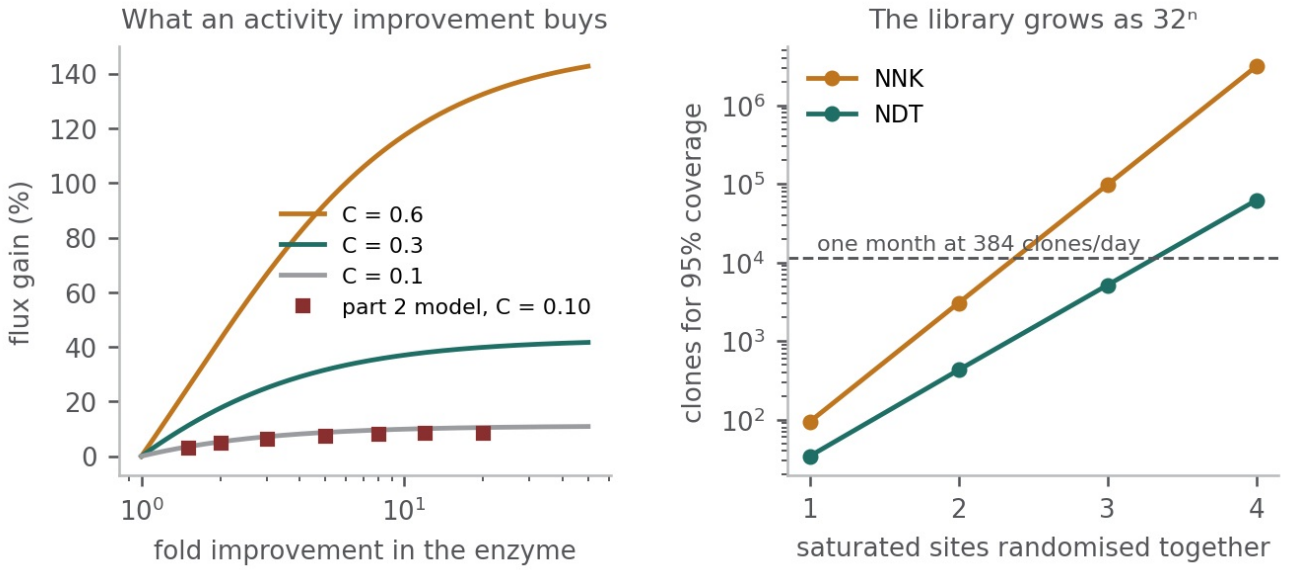


Figure 30.1 Left: flux gain against fold improvement of one enzyme for three control coefficients, from the series-resistance formula, with the full nonlinear model of part 2 chapter 16 overlaid for its C = 0.10 step. Right: clones for 95% coverage of a saturated library against the number of sites randomised together, for NNK and NDT codons, with a month at an assumed 384 clones a day marked. Computed for this module.

The sentence the week asks for, with numbers: *I would engineer enzyme X because it holds 35% of the control, and a ten-fold improvement in its activity would raise flux by 46%, with a ceiling of 54% whatever I do.* If the enzyme held 5% instead, the sentence reads: *a ten-fold improvement buys 4.7% and the ceiling is 5.3%, so I would not* — and the next question is what to check instead, in the order the tracker lists: expression and solubility of the enzyme, cofactor supply, a competing pathway draining its substrate. Those are usually where the real control sits.

An assay you can trust, and a number that says so

Design the assay for one enzyme before anything else, because a library is only as good as its assay: a named substrate, a detection method with a linear range wide enough to cover wild type and the improved variants, initial rates rather than endpoints (part 1 chapter 3, and chapter 22's warning that an endpoint can mislead), a blank, a positive control and a negative control.

The quality of a screening assay has a standard single number, the Z′ factor, from the means and standard deviations of positive and negative controls:

```
Z′ = 1 − 3·(σ_pos + σ_neg) / |μ_pos − μ_neg|
```

Above 0.5 is an excellent assay, 0 to 0.5 is marginal, and below 0 means the two control distributions overlap and the assay cannot separate them. Two constructed examples, since the tracker's data are its own. Assay A: positives 1,000 ± 60, negatives 100 ± 40, so Z′ = 1 − 3(100)/900 = 0.67 — usable for screening. Assay B: positives 1,000 ± 180, negatives 400 ± 150, so Z′ = 1 − 3(330)/600 = −0.65 — the distributions overlap, and no number of replicates fixes an assay whose controls cannot be told apart without improving its precision or its signal. What it implies for a hit, connecting to part 4 chapter 50: you need the calibration, the linear range and the recovery in the real matrix before a number from the assay is a measurement.

Library size, and what a screen can actually find

Saturating a position means randomising its codon. NNK (N = any base, K = G or T) encodes all twenty amino acids with 32 codons and one stop; NDT (D = A, G or T) encodes twelve amino acids with 12 codons and no stop, giving a smaller, chemically balanced alphabet. Clones for 95% coverage of V codon variants at s sites is $\ln(0.05)/\ln(1 - 1/V^s)$, about $3 \cdot V^s$:

Sites	NNK variants	NNK clones	NDT variants	NDT clones
1	32	94	12	34
2	1,024	3,066	144	430
3	32,768	98,163	1,728	5,175

NNK counts are codon-level; because several amino acids have two or three NNK codons, coverage at the amino-acid level needs fewer clones, so these are conservative.

At an assumed 384 clones a day, two NNK sites take eight days, three NNK sites take 256 days, and three NDT sites take 13.5 days. That is the practical reason not to randomise three sites at once with NNK: the library is twenty times the size of the two-site one and well beyond a month's screening. The standard strategy is **iterative saturation mutagenesis**: randomise one or two sites, take the best, make it the template for the next sites, and repeat — which turns a multiplicative search space into an additive one, at the cost of missing combinations whose individual members are not improvements. Choose the sites first from the structure or model of chapter 29: positions within a few ångströms of the substrate, cofactor or the entrance of the channel.

Then the false-hit arithmetic that part 2 chapter 17 introduced, applied to a 10,000-clone screen. If activity is normally distributed among clones that are not truly improved, the one-sided number of false hits above a threshold is $10{,}000 \times (1 - \Phi(z))$: **228 at two standard deviations, 13.5 at three, and 0.3 at four.** A threshold of three standard deviations therefore returns about fourteen impostors, and the first thing to do with any hit list is re-test it in triplicate; the second is to ask how many of the survivors would be expected by chance. That is why a screen's hit rate must be reported with its confirmation rate.

Cytochrome P450 enzymes, and the three limits that are not the enzyme

The tracker's two papers are a good pair. Tyrian purple, a dibrominated indigoid, was produced from tryptophan in *E. coli* with a pathway involving an oxidising enzyme step; the CYP105D18 work engineers a peroxygenase to produce indigo. What they illustrate is what a P450 needs: electrons, delivered either as NADPH through a reductase partner — a two-component system with its own stoichiometry — or, in a peroxygenase mode, directly from hydrogen peroxide, which removes the redox partner and adds a peroxide problem. Both need haem.

Three things that limit the titer, each separable from the enzyme's intrinsic activity. **Electron supply**: the NADPH or the reductase is short, or the reductase-to-P450 ratio is wrong. **Haem supply**: apo-P450 is inactive, and haem is expensive and tightly regulated, so adding the haem precursor δ-aminolevulinic acid to the medium is both a diagnostic and a fix. And **the oxidant** in the peroxygenase case: peroxide inactivates the enzyme and damages the cell, so there is an optimum concentration and a delivery problem — a slow, steady supply rather than a bolus. A fourth, product and intermediate toxicity or insolubility, is the case for the removal strategies of part 3.

What can go wrong between a plant P450 gene and a working enzyme, in the order you would check it:

- **The N-terminal membrane anchor.** Plant P450s are anchored in the endoplasmic reticulum by a hydrophobic N-terminus that misbehaves in bacterial membranes; replacing it with a short, well-behaved anchor, or truncating it, often makes the difference between an aggregated protein and a functional one.
- **The reductase partner**, expressed at a matched ratio and in the right membrane context.
- **Haem supply**, as above.
- **Codon use** — especially of the first codons, where AT-rich, low-structure sequences raise initiation, as part 1 chapter 8 explained.
- **The host.** Yeast has an endoplasmic reticulum and native reductase machinery, which is why it is often the easier host for a plant P450, as part 6 chapter 58 said.

To tell an enzyme limit from a supply limit, **measure active P450 directly**: the reduced carbon-monoxide difference spectrum has a peak at 450 nm for properly folded, haem-bound P450 and at 420 nm for the inactive form, and its amplitude quantifies the functional enzyme per milligram of protein. Then run a lysate assay with saturating NADPH, reductase and substrate. If the lysate activity under those conditions is high and the in vivo flux is low, the limit is supply — electrons, haem or substrate — and engineering the enzyme would be wasted. If the lysate is low even with everything in excess, the enzyme itself, or its folding, is the problem.

Getting a heterologous enzyme to fold

An enzyme that is made and inactive — a strong band, no activity — has a triage, in order of cheapness. Run **soluble against insoluble fractions** to see which; then **lower the temperature and the inducer**, which slows synthesis and gives folding time; then try a **solubility tag** — maltose-binding protein, NusA or SUMO, which are large, soluble fusion partners that raise the yield of soluble protein, with the cost that the tag can interfere with activity or must be cleaved; then **co-express a chaperone**, GroEL–GroES or DnaK–DnaJ–GrpE. The tracker's instruction is to change one thing at a time, and for a triage — where the aim is to find which factor matters — that is right. Once you know which matter, the combination should be optimised by a small factorial, because temperature and inducer interact: a lower temperature tolerates a higher inducer, and testing one at a time hides that. One-at-a-time to diagnose, factorial to optimise.

On codon optimisation: replacing rare codons with the host's most frequent ones raises translation speed, and speed is not always an advantage. Rare codon clusters appear to slow the ribosome at points that help a nascent chain fold co-translationally, so uniformly fast translation can worsen folding. **Codon harmonisation** — matching the pattern of fast and slow codons to the source organism's — is a hypothesis about that, and the way to treat both it and "optimise the codons" is as constructs to be compared: make the native, the optimised and the harmonised versions, express them under identical conditions, and measure soluble active protein, not total protein.

From a hit to a result

A hit from a screen is a hypothesis about one genotype and has five steps to becoming a result. **Purify** the variant and the parent and measure k_{cat} and K_M in vitro — the screen measured something correlated with activity, and this measures activity. **Check expression and solubility**, because an apparent improvement in a lysate screen is often more enzyme, not a better enzyme. **Return it to the host** in the pathway context. **Measure titer against the parent**, with the parent expressed at the same level — otherwise a variant that is merely more highly expressed passes as improved. And include the controls that stop a screen artefact passing as an improvement: an empty-vector strain, the parent assayed alongside the variant on every plate, and a second assay that uses a different detection principle.

Series-resistance gain	`J_new/J = 1/(1 − C(1 − 1/m)); ceiling 1/(1 − C)`
Z′ factor	`1 − 3(σ_p + σ_n)/|μ_p − μ_n|; >0.5 excellent, <0 unusable`
Coverage	`clones = ln(0.05)/ln(1 − 1/V^s) ≈ 3V^s; NNK V = 32, NDT V = 12`
False hits	`N·(1 − Φ(z)): 228, 13.5, 0.3 at z = 2, 3, 4 for N = 10⁴`
Active P450	`CO difference spectrum: 450 nm active, 420 nm inactive`
Folding triage	`fractions → temperature and inducer → tag → chaperone; one at a time, then factorial`

Your pathway's rate-limiting-looking enzyme has a K_M ten-fold above the substrate concentration and a control coefficient you estimate at 0.15. A collaborator proposes a three-site NNK saturation library. What do you say?

The first number settles it. At C = 0.15 even an infinitely improved enzyme raises flux by at most 1/(1 − 0.15) = 18%, and a realistic ten-fold gain gives about 16%. A three-site NNK library needs about 98,000 clones — 256 days at 384 a day — to buy a gain that is smaller than the run-to-run variation of many fermentations. I would not run it.

What I would do instead: with the K_M ten-fold above the substrate concentration the enzyme is operating in its first-order regime, where $v \approx (V_{max}/K_M)\cdot[S]$ and the rate is proportional to both enzyme amount and substrate. So *express more of it*, which raises V_{max} in proportion and costs a promoter change rather than a campaign, and *raise the substrate pool* by improving the upstream step. If the enzyme is then still limiting — the control coefficient will tell — a one-site NNK library at the single residue lining the substrate entrance is a week's work and a reasonable test. The general rule is that a campaign should be proportionate to the ceiling it can reach, and the ceiling is a number you can compute before you start.

Three things people get wrong

- ‘A ten-fold better enzyme gives a ten-fold better pathway.’ It gives 1/(1 − C(1 − 0.1)) — 4.7% at C = 0.05 and a ceiling of 5.3%.
- ‘A band means the enzyme is there, so the problem is activity.’ A P450 band can be apo, misfolded or the 420 nm form; measure the CO spectrum before engineering anything.
- ‘Optimise the codons.’ Fast translation can worsen folding. Compare native, optimised and harmonised versions and measure soluble active protein.

Self-check — answer on paper first

1. An enzyme holds C = 0.4 of the control. What flux gain does a five-fold activity increase give, and what is the ceiling?

1/(1 − 0.4 × (1 − 1/5)) = 1/(1 − 0.32) = 1.47, so +47%. The ceiling is 1/(1 − 0.4) = 1.67, +67%. So five-fold captures 70% of what any improvement could ever deliver, and a further ten-fold beyond that would add little — which is the diminishing returns that part 2 chapter 15 showed from the other side.

2. Your assay has positives at 800 ± 50 and negatives at 300 ± 60. Compute Z′ and say whether you would screen 5,000 clones with it.

Z′ = 1 − 3(50 + 60)/500 = 1 − 0.66 = 0.34, marginal. A hit would have to sit well outside the negative distribution to be called with confidence, so I would expect a hit list rich in false positives and would build in a confirmatory re-screen in triplicate; or improve the assay first — reducing σ by more replicates per well, or widening the separation of the means — because moving from 0.34 to above 0.5 is worth more than screening more clones with a noisy readout.

3. A plant P450 gives a strong band in *E. coli* and a CO difference spectrum with a peak at 420 nm only. What does that tell you and what next?

That the protein is present but not as functional haem-bound P450 — it is the inactive form, or apoprotein with damaged or absent haem. So engineering the enzyme's catalytic residues is premature. Supplement δ-aminolevulinic acid or overexpress the haem pathway, lower the expression temperature and inducer, replace the N-terminal anchor, and re-measure the spectrum: the target is a peak at 450 nm with an amplitude that scales with expression. Only then does an activity assay mean anything.

What to open a book for. Cornish-Bowden's *Fundamentals of Enzyme Kinetics* for what a kinetic constant does and does not tell you; Zhang, Chung & Oldenburg, J. Biomol. Screen. (1999) for the Z′ factor, which is short; the two P450 papers the tracker names for what each enzyme needs; and the literature searches on directed evolution library design, solubility tags and chaperone co-expression. Part 2 chapter 15 is the chapter to read beside the first section.

Streptomyces and natural-product pathway engineering

A natural-product cluster is a pathway you do not have to design and still have to clone, express, feed and sell. The arithmetic of each of those four steps is available before the first experiment, and one of them — cloning a large cluster whole — is worse than intuition suggests.

From cluster to compound, balanced

The tracker starts at the cleanest case: the 6-deoxyerythronolide B synthase, the polyketide synthase that builds the macrolactone core of erythromycin. One propionyl-CoA starter unit and six methylmalonyl-CoA extender units are joined by the assembly line, with reduction steps consuming NADPH. Write it out and check the elements, which is the problem the tracker sets:

```
propionyl-CoA + 6 methylmalonyl-CoA + 6 NADPH + 6 H⁺
       →  6-deoxyerythronolide B (C₂₁H₃₈O₆) + 6 CO₂ + 7 CoA + 6 NADP⁺ + H₂O

carbon   3 + 6×4 = 27   →   21 + 6 = 27
hydrogen 5 + 30 + 12 = 47  →   38 + 7 + 2 = 47        (12 H from six NADPH + six H⁺)
oxygen   1 + 18 = 19   →   6 + 12 + 1 = 19            (acyl parts, CoA-thioesters counted separately)
```

It balances by element count. Three things are worth reading from it. Each methylmalonyl-CoA loses a CO_2 in the condensation, which is why carbon in the product is 3 + 6 × 3 = 21. The single water is released by the one dehydratase in the assembly line. And note the formula: 6-dEB is $C_{21}H_{38}O_6$, molar mass 386.5 g/mol; it is easy to confuse with erythronolide B, which carries one more oxygen and is $C_{21}H_{38}O_7$.

Now the supply bill, which is the reason a precursor or cofactor limit shows up as low titer and not as a failed enzyme. Seven propionyl-derived units go into each molecule — the starter and, via carboxylation, the six extenders — so per gram of 6-dEB:

Titer	Product	Propionyl-CoA units	NADPH	ATP for carboxylation
0.1 g/L	0.26 mmol/L	1.8 mmol/L	1.6 mmol/L	1.6 mmol/L
1 g/L	2.6 mmol/L	18 mmol/L	15.5 mmol/L	15.5 mmol/L
5 g/L	12.9 mmol/L	90 mmol/L	78 mmol/L	78 mmol/L

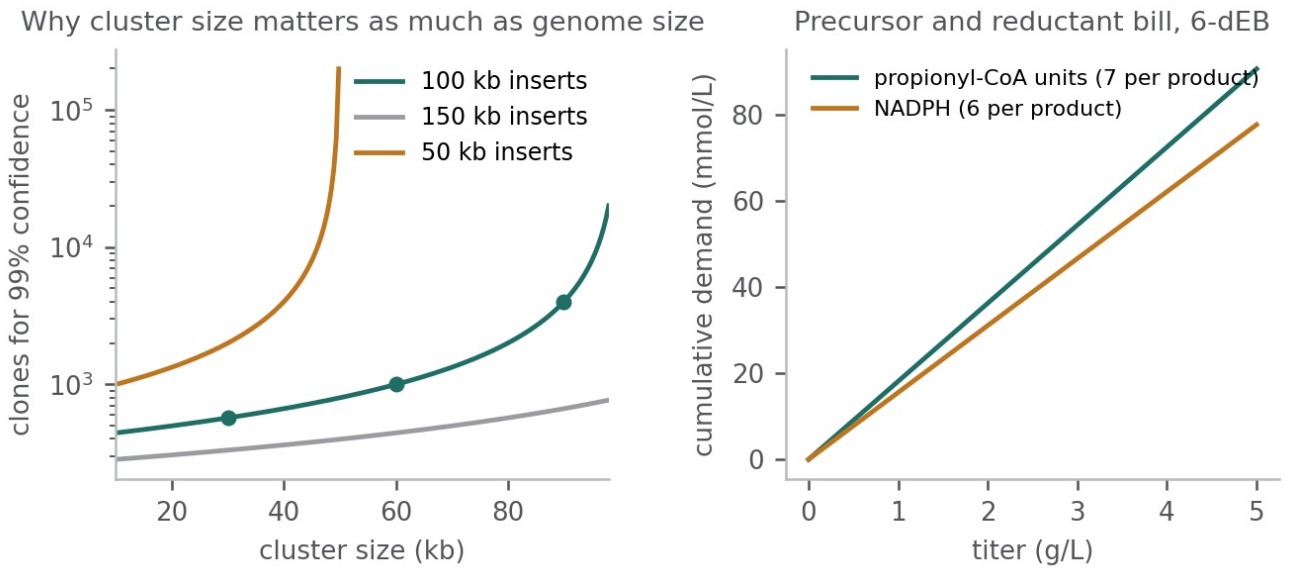


Figure 31.1 Left: clones needed to capture a cluster whole at 99% confidence against cluster size, for three insert sizes, with 30, 60 and 90 kb marked for 100 kb inserts from an 8.7 Mb genome. Right: cumulative propionyl-CoA and NADPH demand against 6-dEB titer. Computed for this module; the cloning model assumes random, unbiased fragment positions.

Converted to mass: 7 × 74 g/mol of propionate per 386.5 g of product, so each gram of 6-dEB consumes 1.34 g of propionate-equivalent, and the ceiling is a mass yield of 0.745 g/g on propionate. At 5 g/L the product alone represents

6.7 g/L of propionate — a weak acid that is toxic to growth at gram-per-litre concentrations at low pH (part 3 chapter 45), so the supply cannot simply be dumped in. It comes from the host's own metabolism — methylmalonyl-CoA from succinyl-CoA by a mutase, propionyl-CoA from propionate or amino-acid catabolism — or is fed slowly. And NADPH at 78 mmol/L, set against part 1 chapter 4's conclusion that NADPH is regenerated at a cost in carbon, explains why polyketide titers are limited by supply long before the synthase is saturated.

Genetics: getting into *Streptomyces*

Streptomyces is a filamentous, spore-forming soil bacterium with a large, linear, high-GC chromosome — about 8.7 Mb and 72% GC for the model *S. coelicolor*. Transformation by electroporation is unreliable, so DNA enters by **conjugation from *E. coli***. The standard donor is a methylation-deficient strain (*dam*⁻ *dcm*⁻ *hsdS*⁻) carrying a non-transmissible helper plasmid that supplies the transfer machinery, because *Streptomyces* carries restriction systems that cleave methylated DNA — the Type IV logic of part 4 chapter 52, and the reason the first thing to try against a methylation-sensitive barrier is a methylation-free donor.

Vectors fall into two kinds. **Integrative vectors** carry a phage attachment site and an integrase — the φC31 system is the standard — so they insert at a defined chromosomal site and are stable without selection; they suit a large cluster. **Replicative vectors** are maintained as plasmids and are copy-number-variable, which is a stability problem for a burdensome cluster. The Streptomyces artificial chromosome (SAC) combines a bacterial-artificial-chromosome replicon, which carries inserts of 100 kb and more stably in *E. coli*, with the integrase and transfer functions for *Streptomyces*, which is why the tracker's pikromycin and neotetrafibricin papers use that kind of system for clusters too large for a conventional plasmid.

Cloning a giant cluster: the arithmetic that surprises

A cluster of size c is captured whole in a clone of insert size i only if the insert's start falls in a window of length (i − c) along the genome of length G. So for random, unbiased fragment positions, the probability that one clone contains the whole cluster is (i − c)/G, and the clones needed for confidence P are ln(1 − P)/ln(1 − (i − c)/G). For 100 kb inserts from an 8.7 Mb genome, at 99% confidence:

Cluster size	P(one clone holds it whole)	Clones for 99%	Genome equivalents
30 kb	0.80%	570	6.6
60 kb	0.46%	999	11.5
90 kb	0.11%	4,004	46

The pattern is the result: **cluster size matters as much as genome size.** Capturing a cluster that almost fills the insert is four times harder than capturing a 60 kb one, because the window of acceptable start positions shrinks to 10 kb. A 90 kb cluster needs 46 genome equivalents of library, not the 3 to 5 that a rule of thumb for random clones suggests, and a cluster larger than the insert cannot be captured whole at all and must be reassembled from overlapping clones.

The model flatters reality in three ways: real libraries have cloning bias, since some regions are poorly cloned or toxic in *E. coli*; a GC-rich genome with uneven restriction-site distribution gives uneven fragment positions; and a clone is only useful if it is verified. Verification is by restriction pattern, junction PCR across both cluster boundaries, and sequencing, and each can miss something — a restriction pattern can match a deleted clone, junction PCR confirms ends and not the interior, and sequencing a 60 kb insert is the only complete check. Compare the alternative from part 7 chapter 21: direct cloning by yeast recombination with overlapping fragments amplified from genomic DNA avoids the library and its bias but needs cluster-specific primers and verification of every junction. Try the library first if a tool exists for your cluster's size and host, and the yeast route when the cluster is near the insert limit.

Silent clusters, and expressing them

Genome mining — antiSMASH and similar tools — finds clusters, and most are silent under laboratory conditions: they are not transcribed, because their pathway-specific regulators are off, or transcribed and not made. Four ways to raise expression, in the order of increasing effort: **replace the promoters** in front of the cluster's operons with constitutive or inducible ones, which refactors the cluster away from its native regulation; **activate a regulator**, often a pathway-specific activator such as a member of the SARP family, by overexpressing it; **move the cluster to a heterologous host** engineered for the job — chassis strains in which the native clusters have been deleted to lower background and free precursors; and **add quorum-sensing control**, as the tracker's *S. coelicolor* example does.

To tell a cluster that is not transcribed from one that is transcribed but not made: RT-qPCR on a biosynthetic gene from each operon, with the validated reference of chapter 22, shows whether transcription happens; LC-MS of culture extracts, with the calibration discipline of part 4 chapter 50, shows whether product exists. Not transcribed points to the

regulators and promoters. Transcribed and not made points to precursor supply, an inactive enzyme, post-translational maturation — a carrier-protein modification, for instance — or product degradation or sequestration; the precursor bill above is the first place to look.

Editing, and why a break is harsh

Cas9 works in *Streptomyces*, with the caveats of part 7 chapter 23 amplified. The nuclease is toxic when expressed strongly, so it must be under tight control and the plasmid must be cured afterwards — typically by a temperature-sensitive replicon. A double-strand break is poorly repaired in the absence of a template, so cutting kills and the repair template carries the edit, with homology arms of one to two kilobases rather than the 40 to 50 base pairs of *E. coli* recombineering. The tracker names a 2025 Cas9 paper and a 2023 review for what had to be engineered to make editing work; I have not summarised their specific modifications, so read the abstracts and log them.

The design the week asks for — a knockout with a homology template, controls and verification: arms of 1 to 2 kb flanking the target on either side, cloned into the editing plasmid with a guide placed within the region to be deleted; **controls** a non-targeting guide, no template, and the empty vector; **verification** by PCR with primers outside the arms, which gives a smaller band for the deletion, then sequencing of the junction, and finally phenotype — loss of the product on LC-MS. Because of the harsh break and the multinucleate mycelium, expect mixed colonies, and plan a round of single-spore purification before any measurement.

Process, and what the product must earn

Filamentous growth changes the bioreactor problem of part 5 chapter 57 in a third way. Mycelium grows as dispersed filaments or as pellets, and both are bad for transport. Dispersed mycelium gives a viscous, shear-thinning broth in which oxygen transfer falls steeply and mixing is poor. Pellets of a millimetre or more have an anaerobic core: with oxygen penetrating a hundred micrometres or so at realistic cell loadings — the dead-core arithmetic of part 3 chapter 48 — a pellet much larger than a few hundred micrometres has idle or dying centres. Morphology control is therefore part of the process, by inoculum density, medium additives and agitation, not an afterthought.

Finally the price. Take part 4 chapter 49's cost model with its parameters unchanged — $3M installed per vessel, $120 per m³ of annual downstream throughput, $3M a year of labour, a 10% discount rate over twenty years — and change only what *Streptomyces* forces: a 100 m³ vessel, a seven-day batch with a day of turnaround, an output of 100 tonnes a year, and propionate at an assumed $1.50 per kg at the 0.745 g/g yield, which is $2 of precursor per kilogram of product:

Titer	Vessels	Capital	Feedstock	Fixed	Capital charge	MPSP
0.5 g/L	48	$168M	$2	$97	$197	**$297/kg**
1 g/L	24	$84M	$2	$64	$99	**$164/kg**
2 g/L	12	$42M	$2	$47	$49	**$98/kg**
5 g/L	5	$17M	$2	$37	$20	**$59/kg**

Utilities add a further $0.15 per kilogram in every row and are not shown.

The lesson is the inverse of chapter 54's. For a commodity, feedstock dominates and yield decides; here feedstock is $2 of a $164 price, and **capital and fixed cost, which scale as 1/titer, are 99% of it.** A 1 g/L process must sell at about $160 per kilogram to break even, which is plausible for an antibiotic intermediate or an agrochemical and impossible for anything priced as a chemical. So the right question about a natural-product project is not "what is the yield" but "what titer do we need for the price this molecule can command" — and the answer is the one that sets the strain-engineering target. These are order-of-magnitude figures from a deliberately simple model, with an assumed $1.50 propionate price and a vessel cost taken from a different product class; the shape of the result, not the third digit, is what transfers.

6-dEB balance	propionyl-CoA + 6 mm-CoA + 6 NADPH + 6 H^+ → $C_{21}H_{38}O_6$ + 6 CO_2 + 7 CoA + 6 $NADP^+$ + H_2O
Mass yield on propionate	386.5/(7 × 74.08) = 0.745 g/g; 1.34 g propionate per g product
Demand per g/L	18 mmol/L propionyl units, 15.5 mmol/L NADPH, 15.5 mmol/L ATP
Cluster capture	clones = ln(1 − P)/ln(1 − (i − c)/G); 570, 999, 4,004 for 30, 60, 90 kb
Pellet limit	oxygen penetrates ~100–200 µm; pellets beyond a few hundred µm have a dead core
Price at low titer	capital + fixed ∝ 1/titer; $164/kg at 1 g/L in this model

You have identified a 75 kb cluster in an *S. albus*-like strain with an 8 Mb genome. A collaborator proposes a cosmid library with 40 kb inserts. What do you say?

That the cluster cannot be captured whole in a clone of 40 kb at all: an insert smaller than the cluster contains at most part of it, so the library would deliver the cluster in pieces that have to be reassembled from overlapping clones, with the junction verification that implies. The arithmetic for a sufficiently large insert is the point: with 100 kb inserts, (100 − 75)/8,000 = 0.31% of clones hold the whole cluster, so about ln(0.01)/ln(1 − 0.0031) ≈ 1,470 clones for 99% confidence — roughly 18 genome equivalents — which is a library of a size that is routine in a BAC system and impractical as cosmids.

So the options are a BAC or SAC-type library with inserts comfortably larger than the cluster — which also gives you the integrase and conjugation functions for the next step — or direct cloning by yeast recombination, which avoids the library entirely, at the price of designing and verifying the overlaps. I would check the cluster's boundaries by genome annotation first, since a 75 kb estimate with uncertain ends changes the insert size you need, and then choose the BAC route if a suitable vector is available for the host, since the host-ready vector saves the transfer step. In either case the verification plan is restriction pattern, junction PCR at both boundaries, and full sequencing of the clone that goes forward.

Three things people get wrong

- ‘A 90 kb cluster in 100 kb inserts is just a bit harder than a 60 kb one.’ It is four times as many clones, 46 genome equivalents against 11.5, because the permissible start window shrinks to 10 kb.
- ‘Low polyketide titer means the synthase is the problem.’ Seven propionyl units and six NADPH per molecule make supply the usual limit, long before the enzymes saturate.
- ‘The product is cheap to make, so the price can be low.’ At 1 g/L the cost is dominated by capital and fixed cost, which scale as 1/titer; it needs a high-value molecule to clear.

Self-check — answer on paper first

1. Balance the production of one molecule of 6-dEB from one propionyl-CoA and six methylmalonyl-CoA, and state the NADPH, CO_2 and water.

propionyl-CoA + 6 methylmalonyl-CoA + 6 NADPH + 6 H^+ → $C_{21}H_{38}O_6$ + 6 CO_2 + 7 CoA + 6 $NADP^+$ + H_2O. Carbon is 27 on each side, hydrogen 47, oxygen 19. Six NADPH, six CO_2 lost at the condensations, and one water from the dehydratase. If your hydrogen count disagrees, check you are counting the reductant's two hydrogens per NADPH and that the product is $C_{21}H_{38}O_6$ and not the more oxidised erythronolide B.

2. How many clones for 99% confidence of capturing a 45 kb cluster in 120 kb inserts from a 10 Mb genome?

P(one clone) = (120 − 45)/10,000 = 0.0075; clones = ln(0.01)/ln(1 − 0.0075) = 4.605/0.007528 ≈ 612, which is about 7.3 genome equivalents. Verification still has to follow: both boundaries by PCR and the interior by restriction and sequencing.

3. A cluster gives no product. RT-qPCR shows the biosynthetic genes are expressed at a good level. What are your hypotheses, in order?

Precursor supply first: seven propionyl-derived units and six NADPH per molecule for a polyketide, which a heterologous host may not provide at the rate needed — test by feeding propionate or overexpressing the carboxylase. Then post-translational maturation: carrier proteins need a phosphopantetheinyl transferase to be activated, and a missing one gives a fully expressed, inactive assembly line. Then product degradation or an undetected product — a different compound from the one the LC-MS method looks for, or one below the quantification limit of part 4 chapter 50. Finally an inactive or mis-folded enzyme, tested by lysate activity with excess precursor.

What to open a book for. The IUBMB enzyme entry for 6-deoxyerythronolide-B synthase and Lowry et al.'s in vitro reconstitution for the chemistry; Pyeon et al., Microb. Cell Fact. (2017) for the pikromycin cluster in the artificial-chromosome system and Kim et al., Front. Bioeng. Biotechnol. (2022) for a giant cluster by BAC; the tracker's 2025 and 2023 *Streptomyces* editing sources for what had to be engineered. Part 3 chapter 48 and part 4 chapter 49 are the two chapters to reread beside the process section.

Closing notes to part 8

The six weeks in one page

Week 26. A part's number travels only as a ratio to a reference on the same plate. A Hill fit gave a baseline uncertain by 95 units with the default routine and 7.5 when weighted by the noise model, so dynamic range spanned 15 to 147 in one and 38 to 49 in the other. The operating window is $81^{1/n}$-fold. A sorting gate keeping 70% of hits and 1% of background takes one in 10,000 to 97% in three rounds, retaining 34% of the true hits. A circuit's speed is its loss rate: dilution alone gives the doubling time.

Week 27. A hundred-fold mRNA knockdown needs the decay rate to rise about a hundredfold — a 1.7-second half-life — so layers must stack. A repressor's ceiling is 1/leak. At a branch point a ten-fold knockdown captures 82% of the available gain when the competitor takes half the flux, and half of it when the competitor takes ten times as much.

Week 28. A three-enzyme library of three levels is 27 variants and about 81 clones. In the burden-aware model, everything-high wins up to a burden slope of about 2.4 and enzyme 1 stays high in every regime. A nine-run orthogonal array picked the wrong level for enzyme 3 because of aliasing. In ALE, single mutations are supplied every generation at 10^{10} cells and specific doubles essentially never.

Week 29. A high pLDDT is confidence in the backbone, not the pocket. ProteinMPNN is ligand-blind; LigandMPNN is not. A score correlated at 0.5 with activity enriches the top 1% thirteen-fold and turns a 30,000-clone screen into roughly 2,300.

Week 30. A faster enzyme raises flux by $1/(1 - C(1 - 1/m))$, ceiling $1/(1 - C)$: 4.7% at 10× for C = 0.05. Two NNK sites are eight days at 384 clones a day; three are 256. A 10,000-clone screen at three standard deviations returns 13.5 impostors. Measure active P450 by the CO difference spectrum before engineering it.

Week 31. 6-dEB balances with six NADPH and one water. Seven propionyl units per molecule make supply the limit. Capturing a 90 kb cluster takes 46 genome equivalents against 11.5 for 60 kb. At 1 g/L the price must be about $160 per kilogram, almost all of it capital and fixed cost.

On the numbers in this part

Every figure and table was computed for this module from the assumptions printed beside it, except where a number is explicitly recalled from the literature and marked so — the biosensor ligand assignments, the pLDDT bands, the benchmark correlation range for variant-effect scores and the wild-type mutation rate of about 10^{-10} per base per generation. Those should be verified against primary sources before they carry a decision. The problem data the tracker refers to (I1, O1, L1 to L4) were not available to me, so the biosensor, assay and library examples use constructed data and stated parameters; the methods transfer, the numbers will differ. Four assumptions carry the most weight: the 4% baseline pathway proteome and the burden slope in chapter 28, which move the winning library member; the 20,000 events a second and 384 clones a day throughputs, which set every time quoted; the random unbiased fragment positions in the cluster-capture model, which real libraries violate; and the $3M vessel and $1.50 propionate in chapter 31's price, which set the dollar figures but not the conclusion that capital and fixed cost dominate.

What remains

Written so far: weeks 1 to 31 and 45 to 61 — 48 of the tracker's 66. What is left is weeks 32 to 44 and 62 to 66. The first group is the data and foundry block (genomic integration, RNA-seq, biofoundry workflows: 32 to 34), the modelling-software block (COBRApy, genome-scale reconstruction, the MATLAB toolbox: 35 to 38), and the design-algorithm, machine-learning and fluxomics block with the process weeks (39 to 44). The second is the publishing, proposal and application weeks. I will take them in tracker order, starting with weeks 32 to 34.